Unraveling Microbial Dynamics in Spontaneous Wine Fermentation utilizing multi- omics analysis

Dominic Hildebrand, supervised by Chrats Melkonian

ABSTRACT

Wine fermentation remains a fascinating process where microbes and their interactions shape the final product. This study tracks *Riesling* grapes from 7 different vineyards in the German Pfalz region in their journey to become white *Riesling* wine. The wines have undergone a sterile fermentation, without outside influence and a fermentation in an industrial winery setting to uncover key changes how the fermentation process influences final flavour. To track vinification, samples were taken at different fermentative checkpoints, utilizing infrared fingerprinting and metagenomics, with a metabolomics sample taken at the end of the vinification process with a sensory panel assessing flavour. The multi- omics integration was correlated to final flavour, resulting in numerous factors influencing final experience. These connections were compared to literature, showing overlap in the results. *Burkholderia*, a bacteria widely present in the fermentation processes was linked to decreasing exotic and microbiological flavours, while increasing sour- mouthfeel experiences linked to its antimicrobial properties by literature. Finally, a novel connection has been made, linking the presence of virusses to increase sweet and smokey sensory characteristics while reducing the experienced honey and sourness. This study produces a large list of correlations, with some of the connections highlighted.

Keywords: Riesling, Wine, Microbial interactions, multi-omics, sensory panel, metagenomics, metabolomics, FTIR time- series,

Introduction

Fermentation is one of the earliest applied food- technologies, with traces of brewing wheat or barley beer going back as far as 13.000 BC [1]. As the real cause of fermentation was unknown until the work of Louis Pasteur in the 1860s, the main application of fermentation prior was through spontaneous fermentation [2]. Prior to this, the primary method of fermentation would have been either spontaneous fermentation or (un)intentional back-slopping; the act of using a starter culture of a previous batch. Back-slopping was described in 1680 while studying beer yeast using the Van Leeuwenhoek telescope, mentioning that in the process of spontaneous fermentation of natural sugary juices, a deposit resembling beer yeast can be found with the power of fermenting pure sugar [3].

In recent years, wineries practicing winemaking using spontaneous fermentation have increased in popularity, using micro-organisms native to the environment of the winery and on the must (crushed grape juice) to ferment [4, 5]. Compared to commonly used inoculum fermentation, spontaneous fermentation increases sensory depth and complexity as compared to the standard inoculum fermentation [6, 7]. The micro-organisms change the compounds within the wine, increasing compound diversity and flavour depth [8].

However, the use of spontaneous fermentation comes at the price of not exactly knowing which organisms are present during and after fermentation. Furthermore, mainstream inoculation improve the stability of the final wine product and reduce fermentation time as compared to spontaneous fermentation[9, 10]. Fermentation could be improved by increasing understanding of dynamics within the spontaneous wine fermentation process. This knowledge could lead to more specific, non-spontaneous wine fermentations by inocculating with a cocktail of desired flavour- enhancing microbes. These fermentations could include the local microbes, increasing the depth of flavours, leaving out off flavours producing microbes. By keeping to local microbes, a truly regional product with desirable flavours can be achieved, as has been performed with yoghurt in the past [11].

The current state of the art regarding microbial dynamics in grapes of the Riesling variety in the Pfalz region in Germany is metabarcoding using 16s or ITS, where even fewer used shotgun metagenomics during their project [12]. Advantages of using metabarcoding are the relative ease and price to perform, as not as many reads are required to have a sufficient resolution. There are also many ready- made processing pipelines available. However, metabarcoding does not encapsulate the entire population and suffers from the inherent biases of abundance estimation, is influenced by PCR amplification [13] and barcode gene copy numbers [14]. Shotgun metagenomics sequences the entire population and can, after assembly, also gain insight into the functional potential of the community [15]. Some downsides to metagenomics lie in the increased number of reads required to obtain sufficient resolution as compared to metabarcoding. DNA of already dead bacteria may also be sequenced and be part of the results, generating limited bias, relevant to both barcoding and metagenomics [16]. Shotgun metagenomics become cheaper over time,

therefore the main downsides are the time and expertise required to overcome the challenges within the metagenomics samples and analysis of said metagenomic information to gain worthwhile knowledge [17].

Furthermore, less attempts were made at connecting specific microbes within the microbiome composition to wine chemistry and resulting flavour. As of late, no projects combining metagenomics and metabolomics on grapes of the Riesling variety in the German Pfalz region has been performed. However, the combination of metagenomics and metabolomics has been performed on Chinese goji berry wine [18], Chinese rice wines [19] as well as numerous other products, such as Chinese steamed bread [20] or Korean kimchi [21], providing insight on the activity of microbial communities in food fermentation. Inclusion of a sensory judge panel, where judges grade products on specific sensory variables can add valuable insight. A sensory panel can be further used to validate findings from other derived datasets from the fermentation to make more meaningful connections. Combining metagenomics and metabolomics with a sensory panel has been performed on a ham fermentation [22], linking microbes to flavour compounds supported by sensory taste evaluation resulting in extra evidence for the found connections.

The aim of this project is to shed light on the complex interplay within the wine must fermentation and uncover which factors, biotic and abiotic, influence the final flavour profile. By examining the ratios of microbes present during this spontaneous fermentation and relating them to final flavour to gain more insight, possibly linking fermentation factors such as specific microbes to flavours. This insight can be acted upon in follow up research and aid in creating a more purposeful fermentation, valuable to improving production quality and creating new stable, complex wine flavours.

Prior research uncovered during yeast isolates from different regions explain around 10 percent of variation on wine aroma[23], giving a hypothesis on at least one factor. It was expected most of the final flavours to be most strongly linked to known factors such as the starting sugar content, known metabolites present and starting acidity. Microbes are expected to individually be less strongly linked to final flavours.

To connect the varying factors within a fermentation to corresponding flavours, 4 different types of datasets were generated using grapes originating from 7 vineyards. The same grapes were spontaneously fermented in a sterile environment with 3 biological replicates using same grape batch, same conditions, same time (named 01 - 07, with biological replicate a, b and c). Sterile means only the grape- skin microbiota is present in the fermentation, without environmental interaction. These same grapes were also fermented different winery environments, each winery location is unique (wgt1 - wgt7, abbreviation for 'weingarten'). Some experiments were performed by addition of sulphite and inoculation with commercial yeast strains (sulphite inoculation 15, commercial yeasts inoculation 17 and 18). Fermentation was tracked by measuring fermentation factors, including but not limited to malic and lactic acid and sugar content to form a time series dataset. During set fermentation stages, multiple metagenomic samples was taken and sequenced using shotgun sequencing. At fermentation end, metabolomics samples were taken, targeting numerous metaboloic compounds. The finished wines were scored by a sensory panel, where a variation of flavours were scored.

Results and Discussion

1 Winery fermentations consistantly increase colour, reduce mouthfeel - sour.

To assess how grape origin and fermentation site (sterile vs. winery) affect wine flavour, the expert panel scores (1–10) were first analyzed across 17 sensory characteristics. Principal Component Analysis (PCA) was then used to reveal flavour differences when applying the same sterile treatment to grapes from different origins, in Figure 1. Within the PCA it can be elucidated that the flavours of the biological replicate fermented wines generally group close together. However, there are some grapes origins that differentiate more in flavour than others. The variation of the biological replicates can also vary, from being close together, which are the wines made from grapes from vineyard 03, 04, 05 and 06 to more spread out with the grapes of vineyard 01, 02 and 07. The variation could be explained due to the subtle differences in the fermentation itself or to the differences by judges scoring the wines. Furthermore, grape chemistry factors including but not limited to glucose and fructose or the tart and malic acid content at of the grapes themselves. The fermentation converts precursor metabolites to compounds influencing final properties such as aroma [24].

To compare the wines fermented in a sterile environment as compared to a winery, more PCAs can be utilized Figure 2. Panel a shows all data, where panel b zooms in on the same grapes fermented in a sterile (vineyards: 01, 02, 05) environment versus fermented in a winery (vineyards/winery: 01, 02, 05). When the same grapes are fermented in a winery, the wines move closer to 'sweet' and 'body' while moving away from 'mouthfeel' and 'sour'. This can be cross referenced in panel C, showing the winery fermentations to have a lower level of sour and mouthfeel and a higher level of sweet and body to their sterile contemporaries.

Investigating to possible causes, these effects were related to other research. This reveals non-*Saccharomyces* yeasts increasing the concentration of volatile esters, increasing the wine complexity[25]. Cross referencing Figure 2 C shows an increase in the complex flavours for the winery fermentation with statistically significant increased levels of exotic for wgt5, honey for wgt1 and fruity odour for wgt 1 and 05 as compared to sterile. All statistically significant differences are shown in Table 1.

To determine the general sensory profile of the wines, a boxplot was generated, visualized in Figure 3 A. Individual judges are plotted on the boxplot to identify patterns and outliers. By merging everything together and plotting it out shows the wines have a 3 point spread in interquartile ranges of the boxplot in final scores of mouthfeel, body, and fruity odour, meaning the wines are most similar in those factors. The wines vary more for other factors.

Relating the sensory panel results to a different *Riesling* sensory panel, it can be seen that the found flavours and literature finds a general agreement, with overlapping values fitting in the boxplots of Figure 3

with flavours including but not limited to sweet at 2.5, sour at 7, bitter at 1, which does not overlap, and honey at 3, with equal medians. [26]. The German wine institute (Deutsches Weininstitut) classifies wines as 'sweet' with sugar content from 45 gr / l or higher, semi- sweet at a sugar content between 18 - 45 gr / l, semi- dry 12 or 18 gr / l, based on acidity and dry with a residual sugar content between 4 and 9, depending on the tartaric acid content [27]. Based on this description, and the averages of the wine being a 4, the wines could be considered dry to semi- dry. Yet, as the wgt2 wine is far on the top right of the PCA and is calculated as statistically significantly more sweet as compared to the sterile fermentation, that wine could be considered sweet in flavour when compared to the other wines, even if only compared to its sterile fermentation. Not to be confused with the legal definition.

To gain insight in the sensory panel consistency and reproducibility, a test was performed utilizing the coefficient of variation. This metric describes the relative difference between multiple retests. If two measurements are close, e.g. 5 and 6, the coefficient is low. If the numbers differ more, e.g. 3 and 8, the coeffieient is high. The variation across the different sensory factors is shown in Figure 3 B. It can be seen some flavours are easier to judge, such as 'fruity taste', 'sweet', 'sour', 'body' and 'mouthfeel' with a low coefficient of variation, meaning the same judges gave a similar score during the replicate taste tests. The boxplots containing small whiskers with low median, suggesting these flavours being easier to judge.

The interquantile range of more abstract flavours, such as 'oxidized apple', 'floral', 'honey' and 'microbiological' have a higher median and large interquantile range, suggest those flavours to be harder to judge, as the judges evidently had a harder time to consistently give similar scores for those flavours.

There is also judge 18, as an outlier for the the flavours 'lemon', 'exotic', 'green' and 'microbiological'. Raw data indicates occurrences of when in the first measurement judge 18 scored a '3' and in the second measurement on the same wine scored an '8'. Perhaps the judge is less experienced or it could possibly be due to a genetic disposition, as evidence exists that some genes are linked to have a genetic disposition to have stronger experience in certain types of flavour [28].

The separate sensory values were also compared to other groups using ANOVA to find statistically significant differences between the wines when compared against each other, shown in Figure 4 with statistical significance marked with an X. A possibly reason as to why there are many significant differences in colour, is because of the high impact to colour a different treatment as compared to sterile has, as can be seen in Figure 2 C, showing a strong increase in colour when looking at winery versus their sterile equivalent. Literature links the colour of white wines to be influenced by pyruvic acid or acetaldehyde. Some microbes can interact with those metabolites, being able to bind or break down those pigmentation metabolites, reducing browning colour intensity [29]. However, as sterile fermentations are consistently lower in colour intensity as compared to winery fermentations Table 1, it could also mean metabolites increasing colour are not produced, or enzymes are being synthesized by microbes present in winery fermentations catalyzing compounds within the fermentation in turn causing browning.

Special attention can be given to the highlighted wgt1 in Figure 4, as it uses the same grapes to ferment the wine as the baseline. It can be seen that 01 is significantly different to wgt1 on 5 sensory categories. These are 'colour', 'fruity odour', 'honey', 'mouthfeel' and 'sour'. It can be seen that the colour, fruity odour and honey increase while mouthfeel and sour decrease. This explains the difference as seen in Figure 2 A and B where the 01 bioreplicates are in the middle of the figure with the wgt1 datapoint moved towards the top right of the figure, with as main contributors being the significantly increased factor honey, and significantly decreased factor of mouthfeel and sour, moving the wgt1 datapoint further to the top right.

To compare the different inoculation treatments with sterile fermentation as baseline, Figure 4 was used. Sterile fermentation 01 can be compared to 15, 17 and 18. All use the grapes from vineyard 1. (SIHAFERM™ Element, Eaton) [30] (17a,b) (Fermivin® 4F9, Oenobrands) [31] (18a,b). Sulfur was also experimented with, adding two doses of SO_2, 40 mg/L (15a-c). The sulfur experiment only had one significant difference compared to sterile 01 in the category colour. yeast inoculation experiments had a different fermentation, with 17 having a significantly higher colour and microbiological category, with a significantly lower 'exotic' and 'fruity odour'. The puzzling aspect is that the SHIAFERM is advertised as 'preserving original grape-specific aroma without forming stronger fruit flavours'[30]. However, as compared to sterile fermentation, there is a significant reduction in fruity odour. Perhaps this is the case when this the yeast is inoculated to a winery fermentation, as wgt1 significantly increases the fruity odour. Perhaps in a winery environment, these two effects cancel each other out, causing a preservation of grape- specific aroma. When looking at 18, the FERMIVIN 4F9 is advertised to produce 'fruity wines with tropical fruit notes and a good mouthfeel' [31]. Yet, the sensory panel perceived only a significant difference as compared sterile fermentation on category 'colour'. There was a slight increase in 'mouthfeel', but not significantly so.

When comparing all of the sterile fermentations versus their winery equivalent, we get an overview of sterile versus treatment, shown in Table 1. In which we can see that for many of different treatments, the colour is higher as compared to sterile. For most of the winery fermentations, the resulting mouthfeel and sour is statistically significant reduced. Inoculations in all but the case of the sterile 07 inoculation of SHIAFERM, reduce the exotic flavour as compared to sterile fermentation. Following this pattern, the wgt6T202 is likely an inoculation experiment, which was unexplained by the data.

Concluding, grape origin can alter final wine flavour drastically and when the same grape is fermented in a winery environment it consistently significantly increases colour as compared to sterile fermentation while reducing mouthfeel and sourness. Inoculations significantly reduce exotic flavours in all but one case, while increasing an experienced microbiological flavour. This combined reduces wine complexity in line with literature [6, 7].

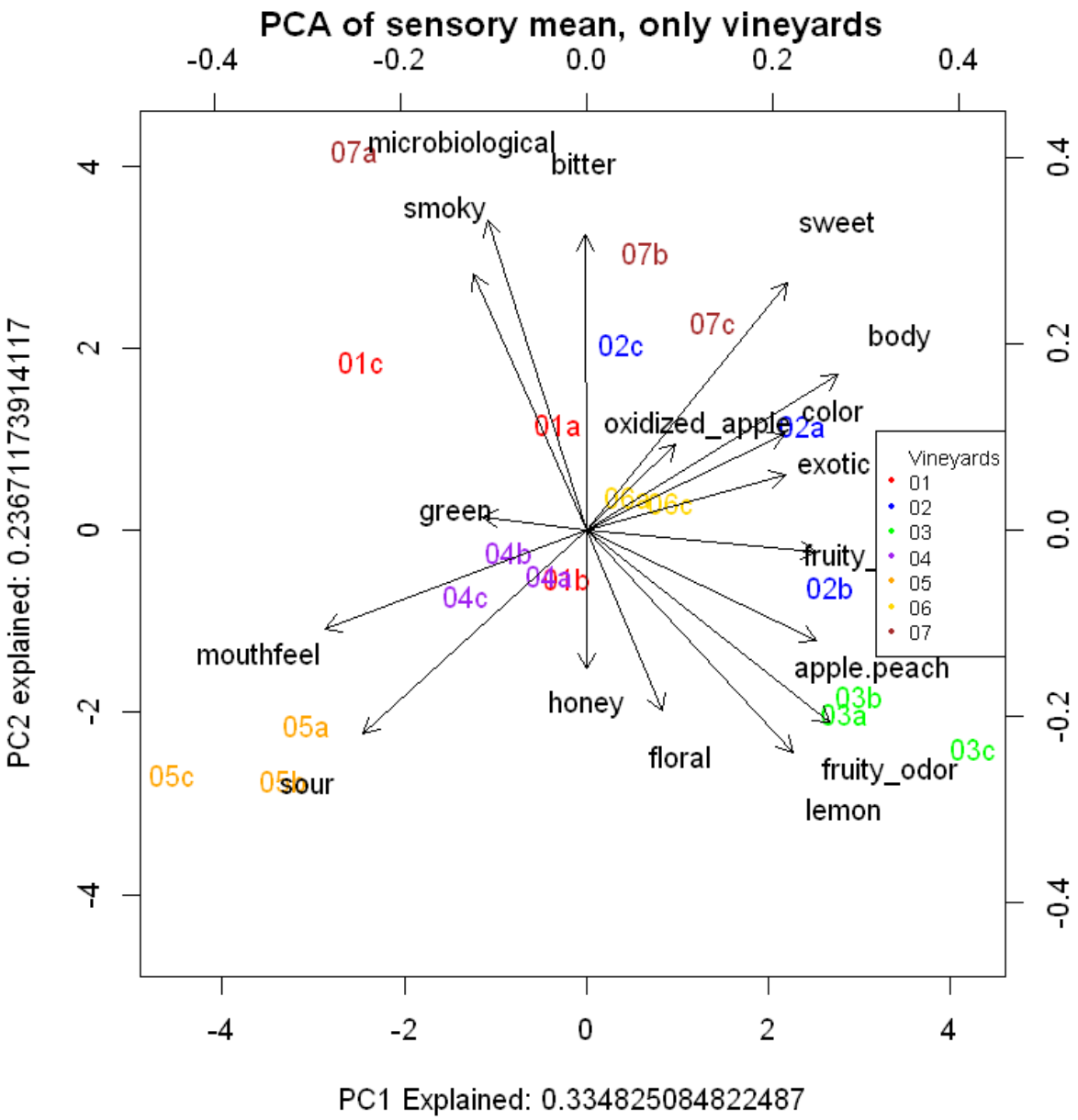


Figure 1. ***Within PCA, biological replicates cluster close together*** PCA of only sterile university fermentation using the sensory data. First two principal components explain 57 percent of all variation within the data. The wines generally group close to each other. Wines 03, 04, 05 and 06a, 06c group close together. Wines 01, 02 and 07 have a larger spread within final flavour.

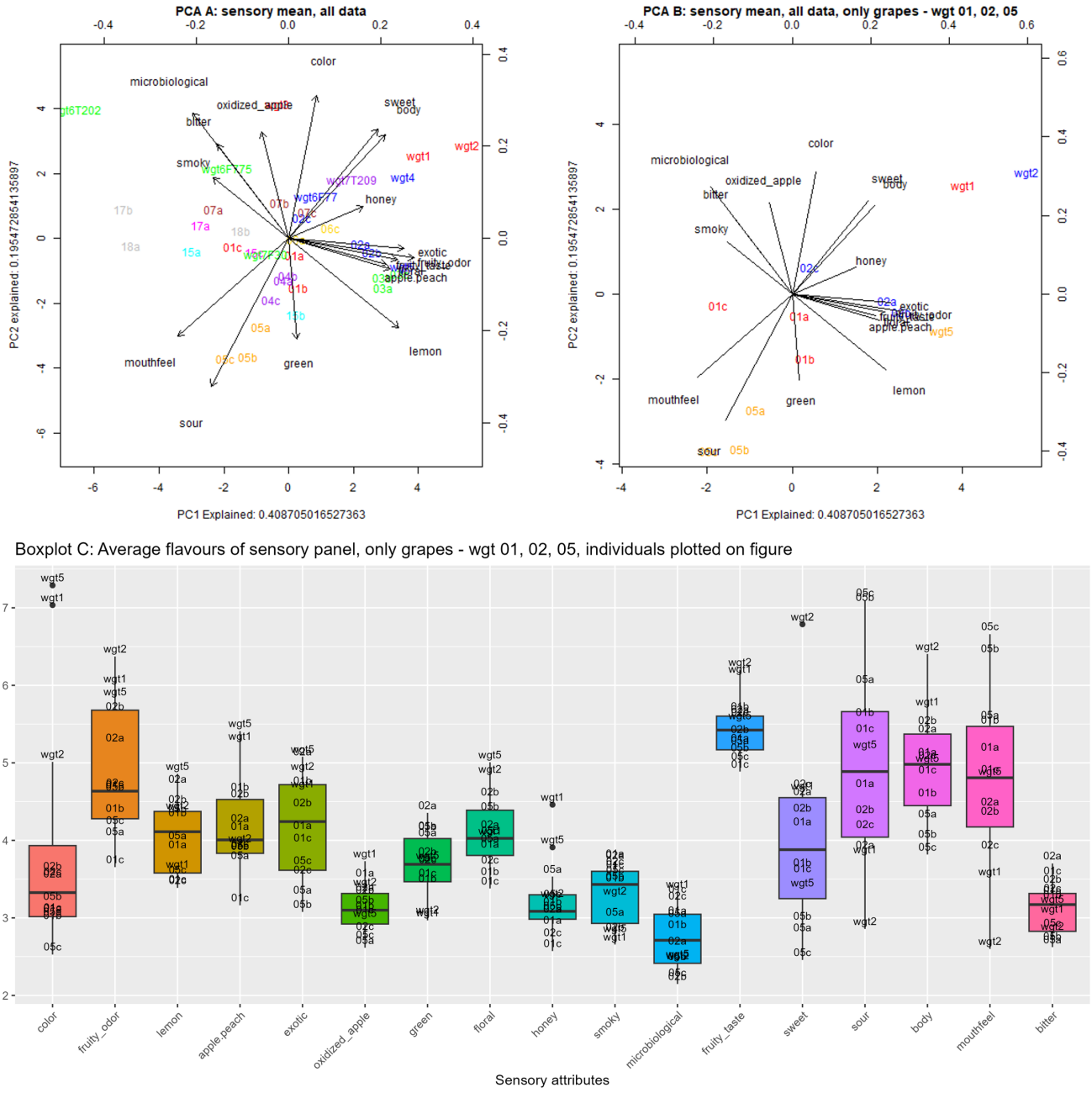


Figure 2. ***Difference in flavour perceived when fermenting in wineries*** (**A**) PCA of all data within the sensory data. First two principal components explain 60 percent of all variation within the data. (**B**) Same data as section A, only visualized sterile 01, 02, 05 and winery wgt 1, 2, 5 as these are the same grapes fermented at a different location. (**C**) Boxplot with each separate datapoint (final wine) plotted on the bars. Each sensory property with judged property for only sterile 01, 02, 05 and winery wgt 1, 2, 5.

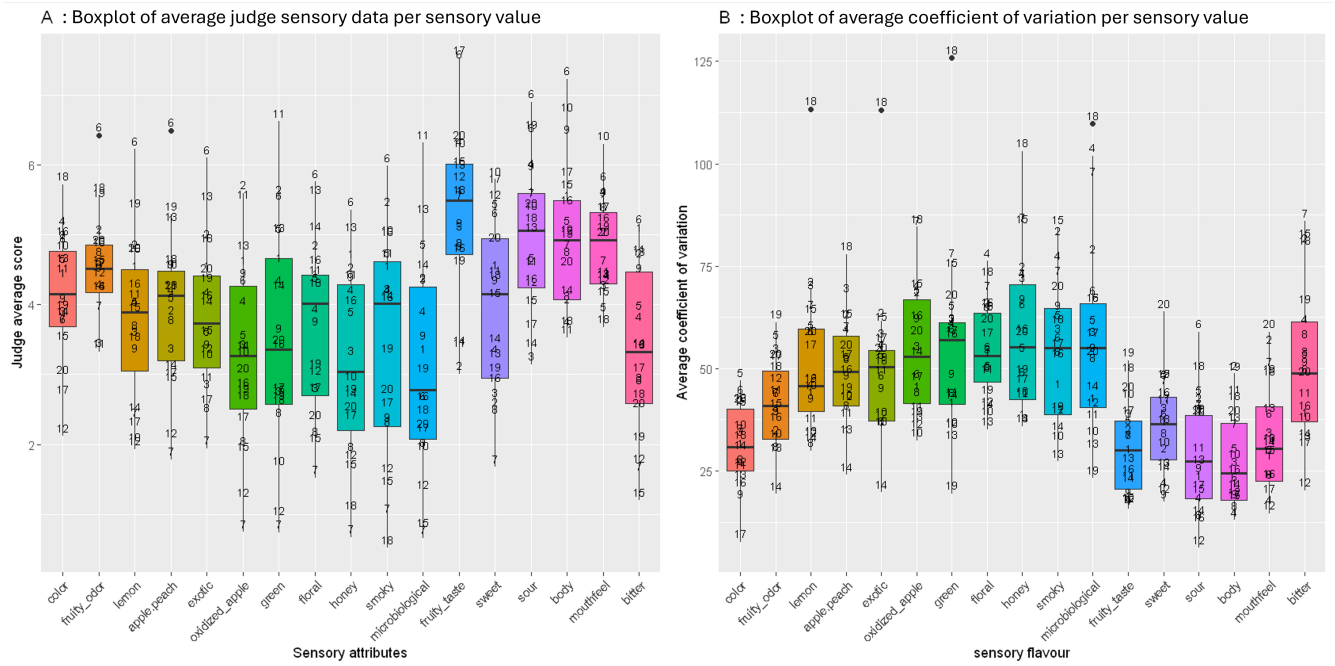


Figure 3. ***Average sensory scores and Coefficient of Variation as perceived by judges with individual judges plotted reveals spread of flavour and difficulty in judging ambiguous flavours.***

(**A**) Average scores of the judges as perceived by the judges. The scores fruity taste, sour, body and mouthfeel have the highest average, which is the middle of stripe within the boxplots. Judges can have different averages and are individually shown in the figure. (**B**) The coefficient of variation is a relative statistic that increases at larger differences between two or more values. The means are calculated and all judges are plotted separately. The boxplots reveal some sensory properties have a relatively high coefficient of variation, where some have a low coefficient of variation. Also visible is that judge 18 is an outlier with sometimes double the average coefficient of variation.

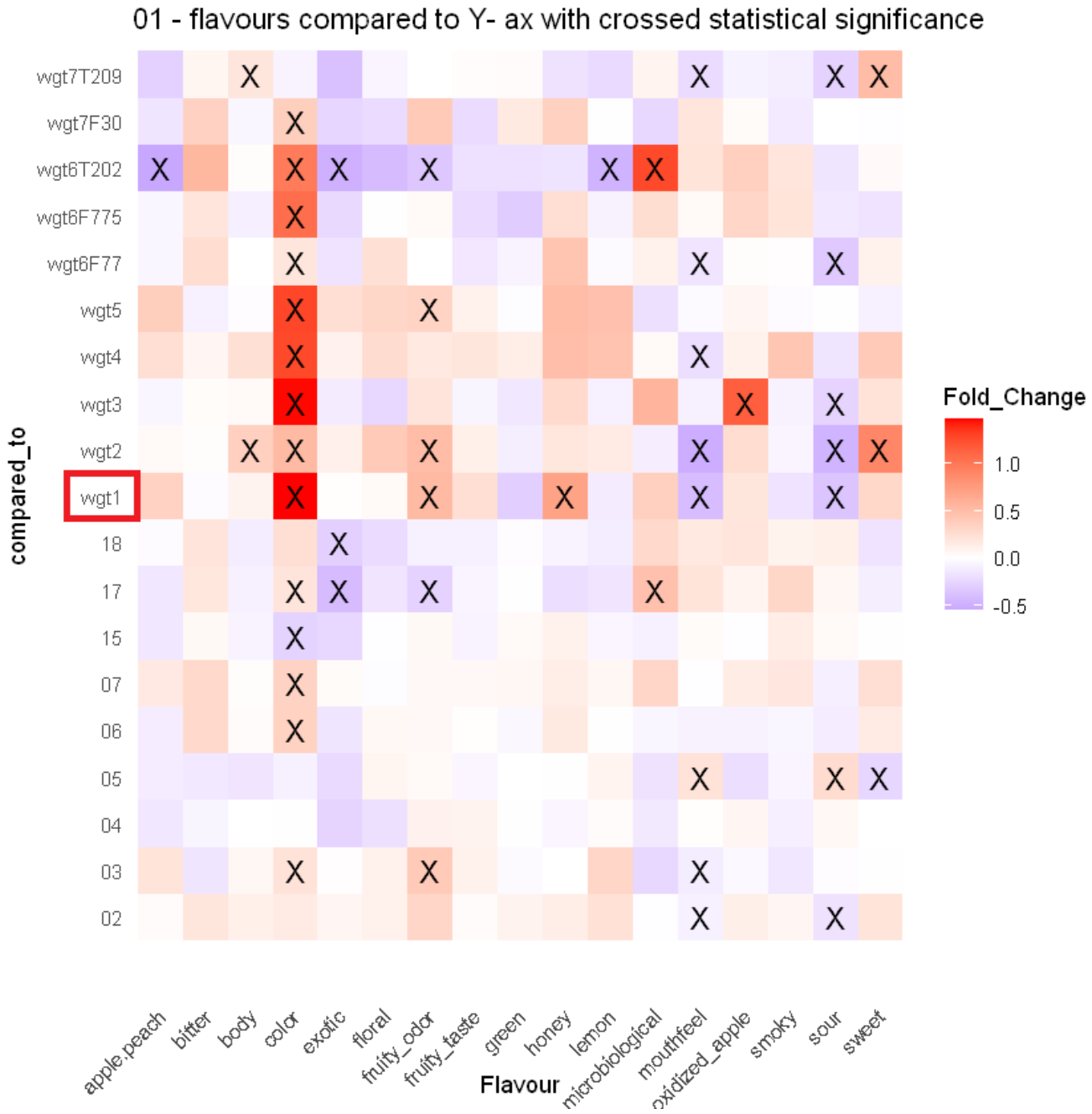


Figure 4. ***Relative change of sensory values with significance marked with an X over all sensory values. Uses merged biological replicates of vineyard 1 as base for comparison. wgt1 is highlighted as this is the sterile vs winery comparison.*** The figure shows the relative difference as a fraction of the target's mean, called 'fold change' or 'fold change minus 1'. It shows a relative value of how the values were scored as compared to the other. A high fold change might indicate a low baseline value, doubling at a fold change of 1. The groups were compared against base sterile fermentation 01 using ANOVA with Tukey multiple testing correction, significant values have a P <0.05 and are marked with an 'X'.

Table 1. Table with only significant differences when applying treatment compared to sterile. H stands for higher and L for lower compared to sterile.

Grapes - Wine	apple.pe	body	bitter	color	exotic	floral	fruity_oc	fruity_ta	green	honey	lemon	microbi	mouthfe	oxidized	smoky	sour	sweet
1 winery				H			H			H			L			L	
1 add sulphite at rack				L													
1 inoc. SIHAFERM				H	L		L					H					
1 inoc. Fermivin					L												
2 winery				H									L				H
3 winery				H								H		H			
4 winery				H									L			L	
5 winery	H			H	H		H						L			L	
6 treat. Unknown F775				H				L									L
6 treat. Unknown F77						H							L			L	
6 treat. Unknown T202	L			H	L						L	H					
7 treat. Unknown F30							H					L					
7 inoc. SIHAFERM													L				
Legend:	Higher	Lower															

2 Metabolomics profiles of wines from wineries differ significantly from those of sterile fermentations

To explore the effects of sterile fermentation versus a winery fermentation and their varying extra experiments on metabolic level, final wine metabolic data was first analyzed using a PCA, shown in Figure 5 A. The first two principal components explain 37% of all data, with a separation on PC2 between sterile versus winery experiments, with the third PC explaining 8.6% of the variation within the data. To further understand what exactly these differences are, the top 10 differences have been plotted in Figure 5 C. showing the top 10 most different metabolites when comparing sterile experiments versus winery experiments. It reveals the primary differences are caused by acetovanillone, ethyl vanillate and c-3-hexenol.

To closer align the goal of finding the difference between sterile and winery fermentation, the primary differences between sterile and winery have also been identified. These results exclude inoculation experiments, winery 3 (missing data), winery 6 and 7 (unclear treatment) to aim specifically at sterile versus winery spontaneous fermentation. The differences are shown in Figure 5 D. The main differences between winery and sterile fermentation are butrytic acid, ethyl acetate and acetovanillone. Cross referencing indicates that for the top 5 metabolites, the wineries have a higher presence in all cases of the top 5 metabolites (Supplementary) Table 5.

To investigate if an overlap between the metabolomics and sensory panel is present, a comparison was made between Figure 5 A and Figure 2 A. In the metabolomics PCA we can see that biological replicates do not cluster together as closely as they do in the sensory panel. For example, the 05c clusters closely to 03a-c, which is on the mouthfeel- sour side for the sensory panel. However, there is overlap in the 18a,b, 17a,b 15a-c inoculation experiments. Those are on the opposite side of the complex mixture of metabolites experienced by the spontaneous winery fermentations. Interestingly enough, wineries consistently have a increased level of butyric acid, a metabolite consistently linked in literature to causing a rancid, microbiological off- flavour. However, not above the threshold many people can measure. The level a sensory panel from literature can measure is at a threshold of 1.73 mg / L [32]. The highest level of butyric acid present in wgt4, at 1.7 mg / L, which is under the set threshold.

To explore the metabolomics using a different unsupervised method, a dendogram was created, shown in Figure 5 B dendogram. When clusters are defined at height 17 for the dendogram the sterile fermentations of 1 through 6 cluster together, with exception of bioreplicate 2b, which was grouped with winery fermentation 1, 2, 4, 5 and 7. Comparing 02b to Figure 5 A PCA, it shows 02b being somewhat further away from the rest of the sterile fermentations. There is overlap in the wgt 6 fermentations, these were grouped separately in both the dendogram and PCA. Winery 7 fermentations were placed with the final winery fermentations, which makes sense as the PCA indicates the wineries have high levels of varying metabolites as compared to sterile. Another part is that the sterile inoculations (17, 18) are

grouped separately from sulphite addition (15) and some sterile fermentations. When comparing the PCA and dendogram together to the flavour in Figure 2 A, we can see the dendogram and metabolite PCA not always overlapping with the final flavours. The metabolomics PCA tells us that wgt5 is very distinct from most of the other samples. However, the flavour PCA tells us that wgt5 is places very close to the bioreplicates 03a-c. This is likely due to metabolomics not capturing all the effects caused on final flavour. Comparing the difference in metabolic landscape to literature, tells us that spontaneous fermentation indeed increases the metabolic landscape of volatiles, specifically highlighting a significant increase isoamyl acetate and ethyl acetate when spontaneously fermented [33], being in agreement of the largest differences found in Figure 5 D.

To investigate metabolites with the largest differences within sterile fermentations, the top 10 metabolites with the highest standard deviation were found and shown in Table 2. From the raw data, it can be seen that biological replicates are not always similar in the way of always having the same metabolites present (supplementary Table 6). For example, the variation in Whiskylactone is driven by metabolic sample 07b with a concentration of 9 ug, with the second highest having a concentration of 1.7 ug. Furthermore, this metabolite is only present in 5 out of the 20 sterile fermentations. Namely, 02c, 03a, 03c, 04a and 07b, hinting that biological replicates can have a different metabolic composition and could explain a part in the subtle variation on flavour. Indeed, this is also visible when looking at the flavour PCA Figure 2 A, where subtle differences in biological replicates are present. The difference in presence - absence could be caused by some microbes being present or absent during fermentation.

In conclusion, the metabolomics between sterile and winery are very different, with many different metabolites present in winery fermentation, with low or no presence in sterile fermentation. The metabolites are not a 1 to 1 comparison to final flavour, meaning some flavour influencing components are not included, translating to a difference in flavour. This makes sense because there are more factors that influence flavour not part of the metabolomics dataset, such as the sugar content, malic and lactic acid and more. Within- sterile fermentations, there seems to be a split present which can be detected using the combination of PCA and Dendogram. The top 10 drivers of variation in sterile fermentation do not overlap with variation of sterile versus winery. For sterile fermentations it also appears that within biological replicates, not always the same metabolites are present.

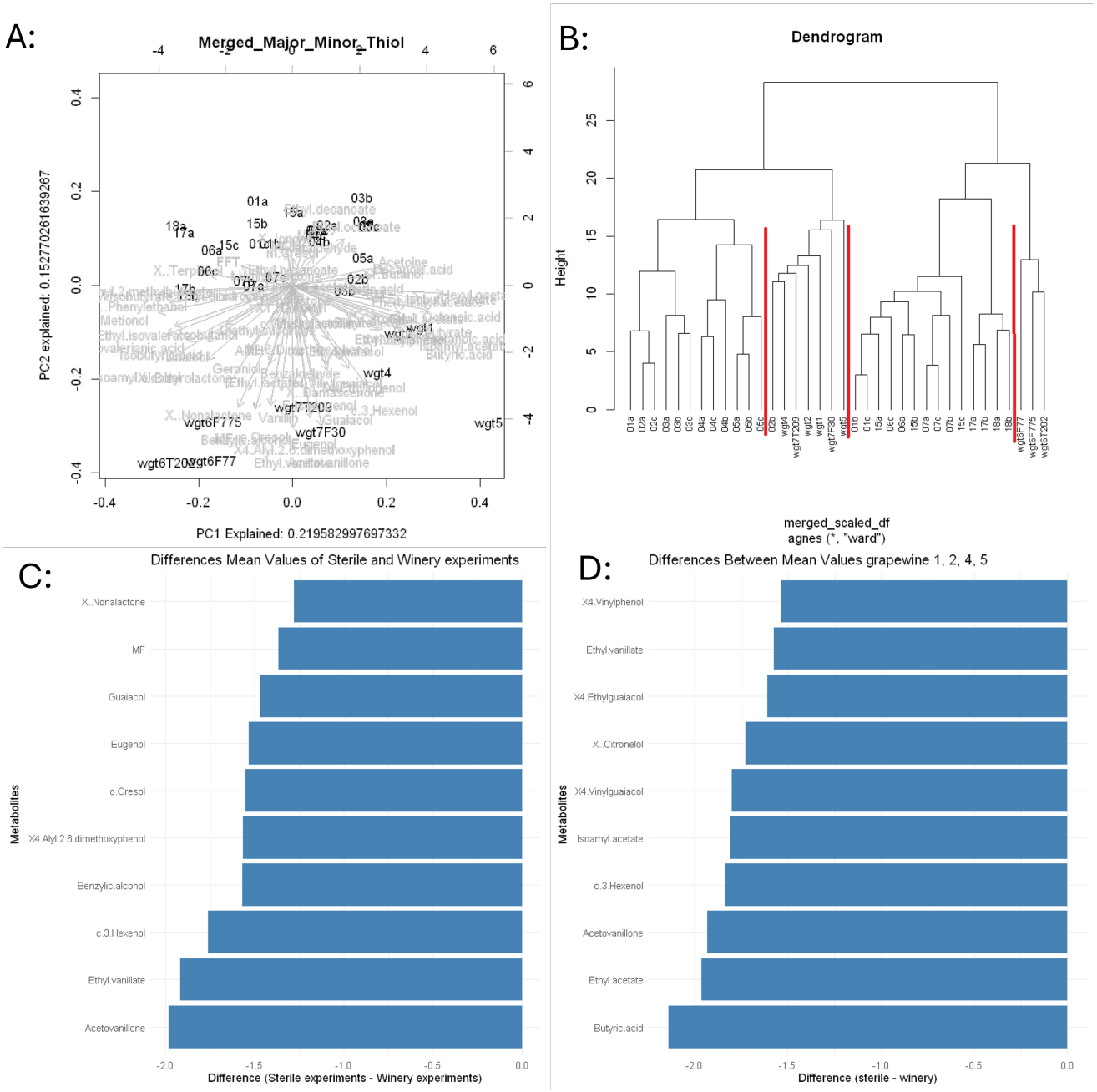


Figure 5. ***Analysis of metabolic data, with the A: PCA, B: Ward's clustered Dendogram, C: Variation sterile experiments vs al winery experiments and D: Variation between grapes and winery.*** (**A**) PCA of all combined metabolomics as retrieved at the end of fermentation. First two PCA explain 37 percent of all variation within data. Arrows show differentiation being caused by the described factor. The third component explains 8.6% of the data. (**B**) Ward's Clustering with an agglomerative coefficient of 0.72. Manually added red lines for increased visibility of separation. Part of the sterile fermentations cluster together. Wgt6 fermentations split from the inoculations of 17 and 18. Sterile 01, 06, 07 group with sulphite addition, separate from other sterile fermentation. (**C**) Top 10 differences between the scaled means of all sterile experiments vs all winery experiments (01 - 18 vs wgt xxx). (**D**) Top 10 differences between the scaled means of only sterile vs winery with only vine - wine 1, 2, 4 and 5.

Table 2. Top 10 variable metabolites within sterile fermentation based on scaled data, calculated using standard deviation. Raw data of the 10 metabolites shown in Table 6.

	top 10 variations
t.Whiskylactone	1,331
Acetaldehyde	1,317
MP	1,292
Acetic.acid	1,160
Decalactone	1,100
Ionone	1,082
1.Hexanol	1,080
MOH	1,078
c.Whiskylactone	1,067
Phenylethyl.acetate	1,017

3 Investigating sugar levels reveals winery fermentation onset delay and possible influences on final sugar levels

To get an indication of the fermentation speed, glucose level over time was inspected, shown in Figure 6 A. Throughout fermentation, microbes convert sugars into alcohol, suggesting a decrease in glucose with an increase in alcohol over time. When comparing Figure 6 A and B, one can indeed find both graphs to have a near perfect inverse relationship between glucose and alcohol levels. This can be confirmed with a correlation of 0.96 Figure 14. Additionally, an delayed increase in alcohol level can be seen for winery fermentation 3 and 5 Figure 6 B, supporting the finding winery 3 and 5 experience a delayed fermentation onset.

To compare glucose and fructose consumption between the different treatments, the fermentations were plotted next to each other in Figure 7 A - F. Fructose consumption is usually incomplete, generally leading to a concentration between 0 - 10 gr / L fructose, mostly classified as dry wine, as the category 'dry' has a limit of 10 gr / L sugar [27]. Fermentations wgt2 and wgt7T209 have the highest level of fructose, above 10 gr / L, leading them to be classified as off- dry. Wgt2 was judged to be the sweetest wine (Figure 2 A and B). Investigating the influence of inoculation compared to the sterile fermentation of 07 - wgt7T209, the sterile fermentation leads to a complete depletion of both glucose and fructose, whereas the inoculation treatment led to a incomplete fructose consumption. This could be explained by some yeast species being glucophylic, causing the yeast to consume glucose while leaving residual fructose [34, 35]. Possibly, the inoculated yeast species was a glucophylic yeast strain and the dominance left no space for a fructophylic species to colonize the fermentation. consume the residual fructose, leading to a sweeter wine, aligning with the sensory PCA Figure 2A. Possibly, the same could be said for the sterile fermentations, if no fructophylic species, yeast or otherwise, are present or the conditions are not right, the residual fructose is unable to be consumed.

To gather insight into the effect of different treatments on the same grape, the unmerged sterile replicate fermentations were plotted of all treatments of the vineyard 1 grapes, as shown in Figure 6 C. It can be that the sterile biological replicates all overlap, meaning fermentation onset for the bioreplicates is similar. Inspecting the winery fermentation, there was a slight delay of several in near- complete glucose consumption as compared to sterile fermentations. The *Saccharomyces* inoculation treatments of 17 and 18, both have a faster fermentation onset and complete glucose consumption compared to sterile. Compared to the sulphite inoculation, having a widely varying fermentation onsets, hinting at a suppressed microbial community. When looking at the glucose consumption of the sulphite inoculations, it can be seen that in all 3 of the replicates there was a varying delay in fermentation onset, as one finished at 20 days after start, ending at the same time as the winery fermentation, one at 40 days after start and a third one ending at 80 days after start. The seen delay in fermentation onset after sulphite addition is reported in prior research, mentioning the strong antimicrobial impact of SO_2 addition even on strong cultural yeast [36]. Addition of sulphites reduced experienced flavours, and low sulphite containing wines containing more vineyard- derived microbes producing wines with more fruity aromas and flavours [37]. It can be seen that this is not the case for our wine, as the addition of sulphites (15) only significantly decreases colour, with a nonsignificant decrease in exotic as compared to sterile, however, this is not compared to winery spontaneous fermentation, as was performed in [37] leading to nonconclusive comparison Figure 4.

Investigating the speed of fermentation led to the conclusion that sterile fermentation have a stable fermentation rate based on consumption of glucose and increased presence of alcohol. Spontaneous fermentation in a winery slows down the sugar to alcohol conversion compared to sterile and inoculation. The inclusion of sulphites on a spontaneous fermentation varyingly delay onset of primary fermentation. The differences in sweetness could potentially be explained by presence or absence of a fructophylic species able to consume fructose. With absence, the wine ends up with a higher concentration of fructose.

4 Time- series data reveal winery fermentations to produce larger concentrations lactic acid

Generally in a wine fermentation, after the primary alcoholic fermentation, the malic acid gets converted to lactic acid in the secondary fermentation by lactic acid bacteria. As such, it is of interest to investigate and compare the levels of malic and lactic acid. The malic versus lactic acid is plotted out in Figure 6 D and E, showing the sterile and winery experiments respectively. Plot E only shows the sterile and vineyard 1 inoculations to reduce clutter. The plots are connected by days. Malic acid always starts on the high end of the fermentation and gets consumed throughout the fermentation, with some exceptions being the presence of rare malic acid creating strains of yeast, generally producing between 0 - 1 or up to 3 gr / L worth of malic acid throughout fermentation[38, 39]. As such, the start data is in all cases the point furthest unconnected dot on the left side of the plot. It can be seen that in most of the fermentations the

malic acid and the lactic acid levels get reduced, in a downward trend, with exception of vineyard 7, which forms an inverse bell curve, signifying an increase in lactic acid is starting to occur. However, literature explains lactic acid is generally created and accumulates, not consumed [40], making the decrease in lactic acid shown odd. The final position of the lactic acid in sterile fermentations is often around 0.2 gr/L, with winery ending up at 0.6 gr/L with exception of wgt3 and wgt6F77.

To compare sterile versus winery, malic to lactic ratios, the winery fermentations have been plotted separately in Figure 6 E. It instantly becomes clear that winery 3 and 6 have a largely increased level of lactic acid near the end of fermentation. Winery 3 also has a strange jump from low malic and high lactic, to low and back again, hinting towards a discrepancy in the data.

To investigate a possible cause as to why grapes from vineyard and winery 05 are much more sour than the other grapes in relation to the time- series data, their content of malic and lactic acid have been investigated. Grapes from vineyard 05 were considered as a more sour- mouthfeel wine, with the sterile fermentation being experienced more sour with a score of 6 - 7 whereas the winery a score of 5 (Figure 2 A, B and C). Both the sterile and winery fermentations keep relatively high levels of malic acid present, with the winery fermentation a slightly higher level of malic acid present at 4 g per litre as compared to 3.5g per litre for the sterile fermentation. In both cases, the level of lactic was low. On the winery side the vineyard 5 fermentation, the data goes into the negative, hinting at measurement errors as negative presence of lactic does not exist. Malic - lactic content show a -0.49 correlation(Figure 14).

To conclude, in the cases of the sterile fermentations, it appears both malic and lactic acid are consumed, generally ending at a level of 0.2 gr / L lactic acid with exception of 07. The winery fermentations end at higher levels of lactic acid, in most cases 0.6 gr / L, with exception of wgt6F77 and wgt3, ending at 1.5 and 2.3 gr / L respectively. Thus, the winery fermentations end up having a higher level of lactic acid.

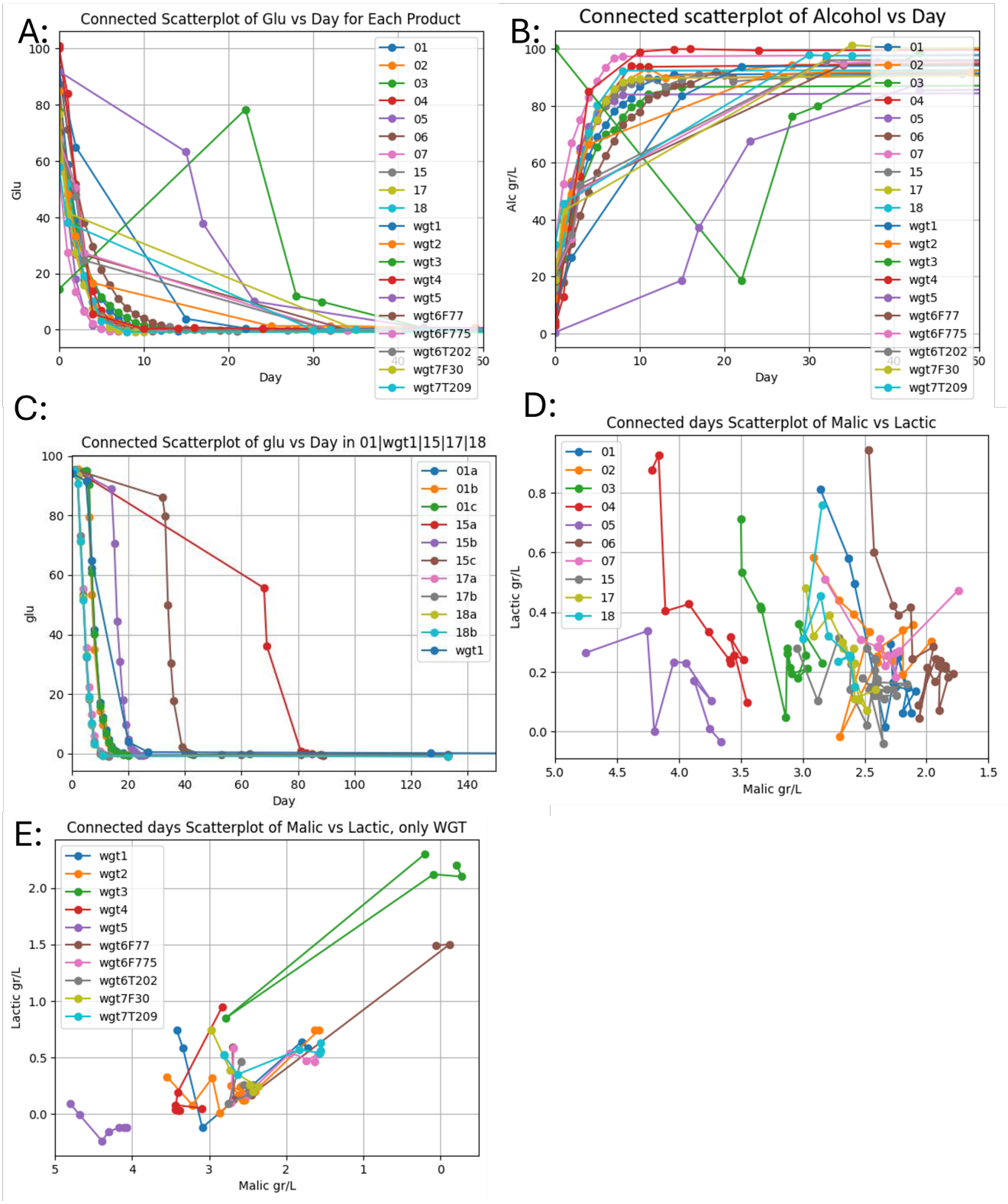


Figure 6. ***All images as related to the time- series analysis, highlighting glucose, alcohol and lactic - malic acid. Merged bioreplicates.*** (**A**) Connected scatterplot of Glucose per Day for each product, Y- ax is grammes per litre. (**B**) Connected scatterplot of alcohol vs Day for each product, Y- ax is grammes per litre. Datapoints have merged bioreplicates. The outliers present are wgt 3 and 5. (**C**) Connected scatterplot of glucose vs day for all the grapes of vineyard 1. Bioreplicates plotted separately. Sulphite inoculation linked to delayed sugar consumption. (**D**) Malic vs Lactic acid level, days connected. Only sterile experiments. Merged bioreplicates. Start with high malic levels, with slight increases in lactic acid level at the end. (**E**) Malic vs Lactic acid level, days connected. Only winery experiements. No bioreplicates for winery fermentations.

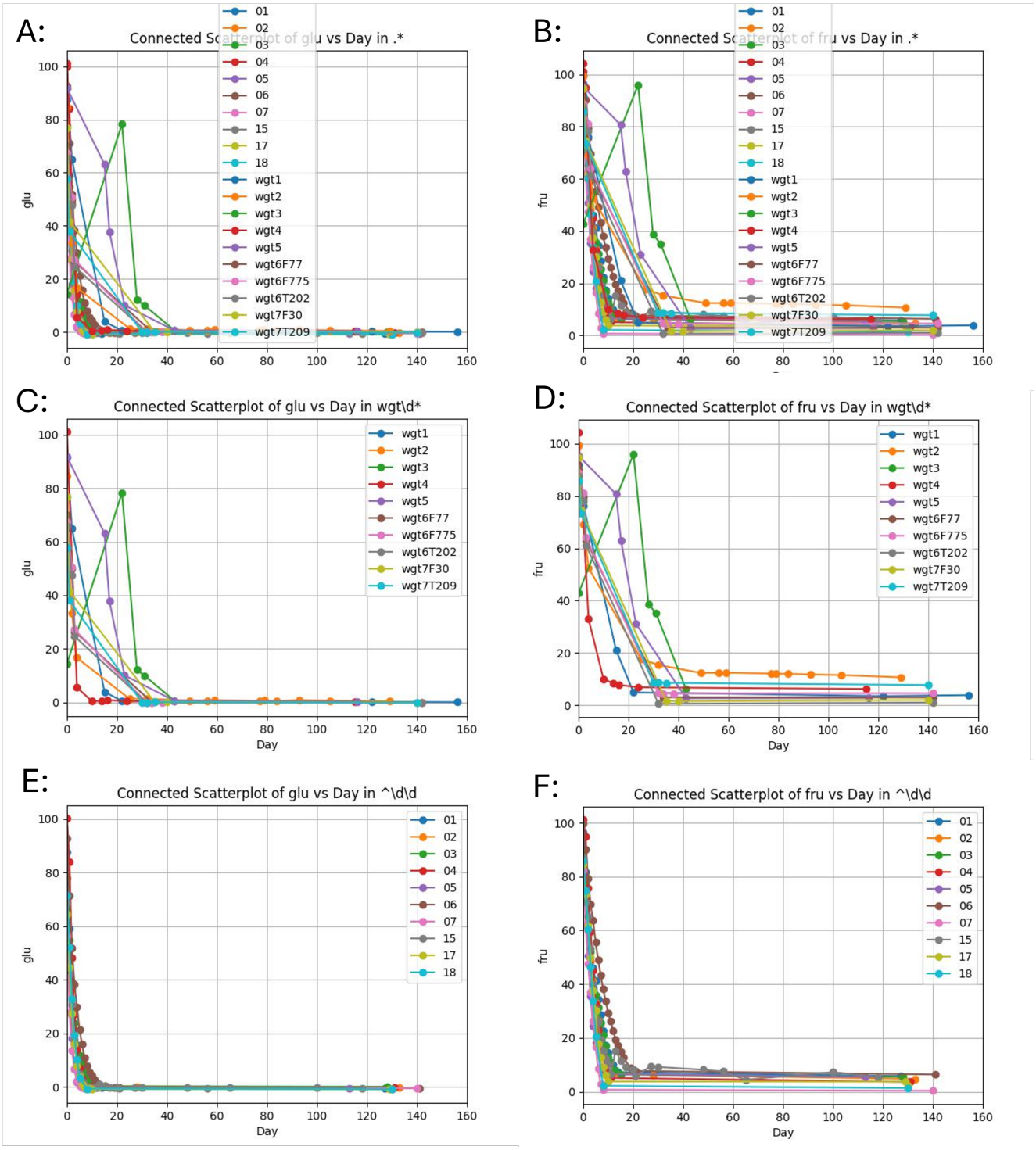


Figure 7. Glucose consumption versus Fructose consumption. Data is from merged averaged bioreplicates and the whole fermentation process shown. Y- ax is glu or fru gr/L. Shows complete fermentation. Residual sugars are primarily fructose. Glucose consumption is complete in all cases. Wgt 7T209 and 2 are the highest level of fructose. Glucose consumption is complete.

(**A**) All data, glucose vs day fermentation. (**B**) All data, fructose vs day fermentation. (**C**) Only wineries, glucose vs day fermentation. (**D**) Only wineries, fructose vs day fermentation. (**E**) Only sterile experiments, glucose vs day fermentation. (**F**) Only sterile experiments, fructose vs day fermentation.

5 Filtering steps remove 48% of reads

To analyse the metagenomics data, an in- house pipeline was created and utilized. One of the methods to remove plant reads is to map against a reference genome. The selected reference genome is the NCBI RefSeq *Vitis Vinifera* highlighted reference [41]. After mapping, we can see that between 20% - 30% of reads were mapped to the reference for most metagenomics files Figure 8 A. The most reads mapped to reference at 73% was from a sample taken at the pressing stage, where the juices are extracted from the grapes. The least mapped to reference were from a library negative sample at 2 percent. The decision was made to extract both forward and reverse read if either mapped to the reference, and as such the mapped percentage can be interpreted as percentage of reads removed from the metagenomics samples. A more detailed overview for each group of samples mapped to reference can be seen in supplementary Figure 15.

To analyse the reference based classification results, there are several things to take into account. Such as the level of classification on the different taxonomic ranks. For the most accurate analysis possible, the lowest rank is be selected with the most assigned reads. The assignment levels can be seen in Figure 8 B. As over 75 percent of all reads could not be assigned to species level, the decision was made to use genus level, as it is only one class higher than species and still has most of the reads present assigned. An interesting observation can be made by looking at the phylum level, which still has on average 30 percent of all reads unassigned. This could indicate not all plant- related reads were removed from the data, possibly due a lacking reference genome, which could not completely remove all reads or to contaminants in the form of human DNA, other plants or other high eukaryotes such as bugs. Due to lack of time this was not further analyzed.

To conclude, mapping on removes around 25% of all reads. Further classification is unable to assign around 30% of the remaining reads, hinting at incomplete filtering or contamination. This means 52.5% of all reads were left over for analysis.

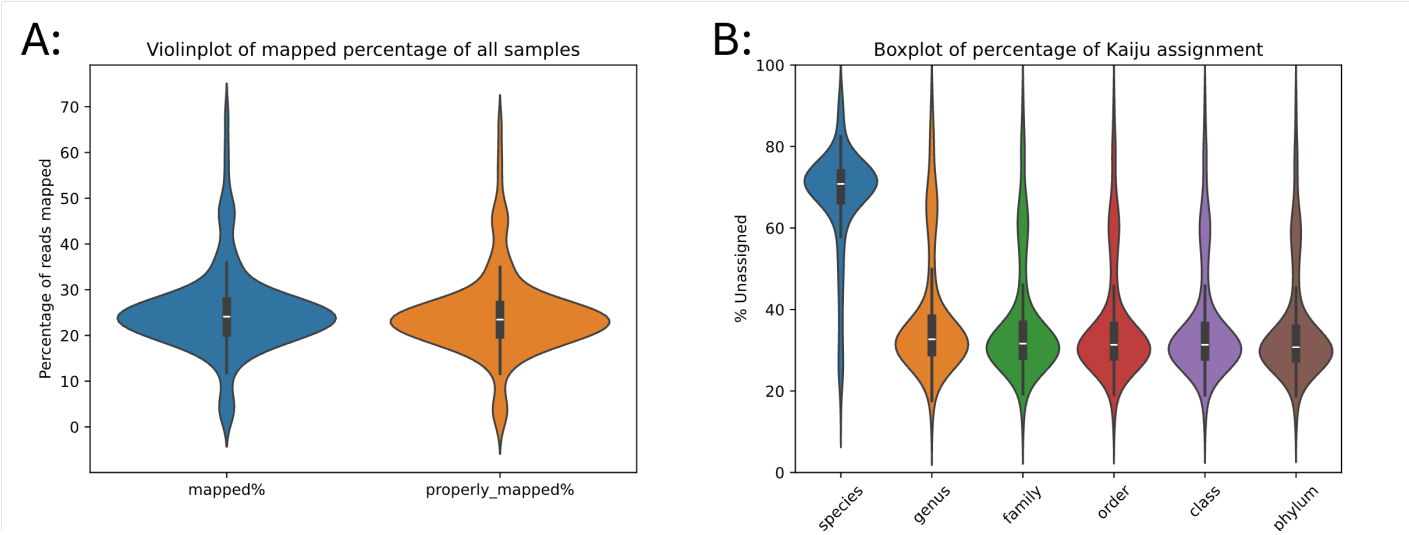


Figure 8. ***Plotting the metagenomics Illumina reads discarded utilizing reference- based mapping reveals a relatively low number of reads removed. Genus has least unclassified reads at lowest rank.*** (**A**) Violinplot indicating percentage of reads removed using reference based mapping on *Vitis Vinifera* [41]. Mapped percentage is the number of reads where at least one of the two reads, either forward or reverse, map to the reference genome. Properly mapped means both forward and reverse read map to the reference genome. The highest amount of reads mapped to reference is 73 percent, the lowest 3 percent. Reads mapped to reference can be interpreted as reads removed from metagenomics sample. More detailed plot available at Figure 15.
(**B**) Violinplot indicating percentage of reads 'unclassified', as nothing matched in the reference library. Species level has unclassified average of just over 75 percent, while genus through to phylum has an unclassified percentage of 34 down to 30 percent.

6 Microbial composition start with high diversity, *Hanseniaspora* often dominates onset, superceded by *Saccharomyces during alcoholic fermentation*

To understand microbial composition over the time of fermentation, the samples have been sorted by timepoint of the fermentation and were placed next to each other in a percentage- based stacked barplot Figure 9 A. From this figure we can learn that during the sedimentation and onset of the wines there is a relatively larger abundance of *Hanseniaspora* as compared to *Saccharomyces*. This can be followed up by inspecting the sterile fermentation onset and sedimentation phase, where it can be seen that during the early press and sedimentation phases, there is a very diverse set of microbes present, from *Aureobastidium* to *Botrytis* and *Pseudomonas* Figure 10 A and B. However, generally, *Hansenisapora* is often abundant during the sedimentation and often dominant during onset of fermentation. In other words, during press and sedimentation phase there presence of a large variety of microbes compared to primary alcoholic fermentation. The raw Alpha Diversity statistics for vineyard and winery 2 provide a Shannon diversity index above 4 in both sedimentation and onset phases for the biological replicates, with exception of sample 02b sedimentation only having a Shannon Diversity score of 3.4 (supplementary Table 7). The Shannon Diversity is an indication of specie richness. The lower the score is, the higher the dominance of one or few species. The samples of the main alcoholic fermentation have a Shannon Diversity score of around or under 1, indicating dominance of one or few species. Looking at the Ace and Chao1 diversity scores, both estimators for the number of species present, show that during press and onset there is a higher estimated diversity of species present as compared to the main alcoholic fermentation. There can also be looked at the Simpson Evenness, which indicates that during press, sedimentation and onset, there is a relatively higher evenness at 0.8, meaning a high diversity where the rest of the fermentation on average has an evenness of 0.27, indicating fewer species present and higher dominance of one or few species.

To compare vineyard and winery 2 against the other vineyards, the average Alpha Diversity statistics were taken and placed in Table 3. Inspecting this table we can see some patterns; at the press and sedimentation stage, a very high percentage of reads is unclassified, 66% and 51% respectively. During more advanced stages of fermentation, this number reduces. Likely this is due to consumption of contaminant DNA by the present microbes, or due to the higher abundance of other microbes in relation to the contaminant. It can be seen that the average Simpson's Evenness (between 0 - 1, higher is more diverse, lower is less diverse) is high for the press and sedimentation phase, namely 0.73 and 0.69 respectively, meaning the microbial field is diverse. During the alcoholic fermentation, there is little diversity present, at hovering between 0.12 and 0.18. After primary alcoholic fermentation, the diversity increases again to 0.25 when racking, hinting at an increase in diversity caused by the lactic acid bacteria gaining ground. Comparing the Shannon Diversity found to literature, it can be seen that grape must shows similar diversity in the press phase as to what is described as 'must', having a median Shannon Diversity of 4 [42]. This

is similar to what was found in vineyard 2 specifically and similar to the calculated average. The high diversity, to the decrease and disappearance of some yeast species to later increase in lactic acid bacteria is discussed in other literature without any alpha diversity connection to this phenomenon [43].

In conclusion, the microbial composition starts very diverse with genera such *Botrytis*, *Aureobastidium* and *Hanseniaspora*, until *Hanseniaspora* often briefly gains dominance during fermentation onset. Often, but not always, *Saccharomyces* succeeds during primary alcoholic fermentation, reducing diversity. After alcoholic fermentation, the diversity increases again driven by lactic acid bacteria.

7 Winery 1 fermentation high presence of *Hanseniaspora* instead of only *Saccharomyces*, flavour influences in line with literature

To investigate patterns, the different fermentation groups were studied. The winery 1 fermentation stood out, shown in Figure 9 B. In the winery, main alcoholic fermentation had high presence by *Hanseniaspora* instead of *Saccharomyces* as seen in other fermentations, like Figure 9 A, showing vineyard and winery 2 experiencing the phenomenon of high *Hanseniaspora* presence on fermentation onset, and getting superceded by *Saccharomyces* during primary alcoholic fermentation. Comparing this to all known sedimentation and onset datapoints, shown in Figure 10a and Figure 10b, it can be seen that in many but not all cases, *Hanseniaspora* is the most present during fermentation onset. An interesting observation can be made that in most of the cases where *Hanseniaspora* is not the most abundant during fermentation onset, that *Hanseniaspora* was the most abundant in the prior sedimentation phase. This is the case for 01a, 01b, 03b, and 07b. This was not true for 06a and 06b, where *Hanseniaspora* was not the most dominant in either sedimentation or onset. Hanseniaspora abundance in fermentation is well documented, where the prevalence of *Hanseniaspora* is attributed to its rapid growth, but losing out to *Saccharomyces* throughout fermentation due the decreased tolerance to ethanol [12, 44, 45].

The biological replicates of Figure 9 A and B, vineyard and winery 1 and 2 follow a similar pattern in both, having grapes with *Aureobastidium* and *Botrytis* present in relatively large numbers, where sedimentation and / or onset is dominated by *Hanseniaspora* and *Saccharomyces* supercedes *Hanseniaspora* during primary alcoholic fermentation. The winery 1 fermentation is unique, as *Hanseniaspora* retains a high presence throughout alcoholic fermentation. Its uniqueness can be confirmed by comparing the the supplementary sterile and winery fermentations Figure 17a, Figure 17b, Figure 17c , Figure 17d, where none of the other fermentations have a degree of *Hanseniaspora*- as high as in winery 1 throughout alcoholic fermentation, making winery 1 an interesting case study. It could be possible that winery 1 contains a uniquely resilient species or strain capable of withstanding the high- stress environment. Several species of *Hanseniaspora* have been identified to have poor or high tolerance to fermentation environment, describing *Hanseniaspora Uvarum* as the most tolerant. [46]. However, as the must is the exact same as the as the sterile fermentation, which did not experience the *Hanseniaspora* presence throughout fermentation.

To account for this, winery 1 samples were investigated at species level, shown in supplementary figure Figure 18, showing the species present in all vineyard and winery 1 fermentations to be *Hanseniaspora Uvarum*, the resilient strain hinting at other microbial interactions or effects to cause the relatively high prevalence of *Hanseniaspora* in winery 1.

When comparing the sensory experience when fermenting sterile vineyard 1 grapes utilizing Figure 2, we can see that the fermentation of 01a-c to wgt1 is not much different from 02a-c to wgt2, where in both cases the winery fermentation is pulled about an equal distance to the sweet- body corner on the top right. This can be cross referenced to Table 1 where difference from 01 to wgt1 is significant in 'colour', 'fruity odour', 'honey', 'mouthfeel' and 'sour', while the significant differences between 02 and wgt2 is 'colour', 'mouthfeel' and 'sweet'. When comparing a *Hanseniaspora* driven fermentation to a *Saccharomyces* driven fermentation, it is mentioned in literature that *Hanseniaspora* increases the level of acetate esters. This increased level in turn gives the wine a positive fruity aroma and honey aroma [47]. This is in line with the findings the difference between fermentation in vineyard 01 sterile versus winery 01, where among the differences is a statistically significant increase in 'fruity odour' and 'honey'.

In conclusion, the winery 1 fermentation is unique due to the retained high presence of *Hanseniaspora*, possessing a strain capable of surviving in a high- stress alcoholic fermentation. Compared to the sensory panel, there is a significant increase in fruity aroma and honey, linked to *Hanseniaspora* through literature.

Table 3. The average Alpha diversity for each of the sterile replicates.

	Average Press	Average Sed	Average 2.5%	Average 4%	Average 8%	Average 11%	Average Rack
unclassified	28951773	13192894	15957572	10099047	12803543	14187726	14480968
sum_counts	43385378	26226378	51199133	32160701	40846246	48194733	50584414
%unclassified	66,30	51,45	32,31	32,47	31,50	30,24	28,24
ShannonDiversity	3,58	3,35	0,76	0,67	0,79	0,72	1,03
AceDiversity	5001,83	4712,80	3555,01	3134,10	3254,29	3686,18	3974,70
Chao1Diversity	5062,60	4769,55	3633,79	3203,59	3324,80	3770,28	4062,55
SimpsonEveness	0,730	0,693	0,168	0,128	0,174	0,149	0,252
GiniDominance	0,99147	0,99213	0,99905	0,99914	0,99911	0,99919	0,99893
GoodsCoverage	0,99991	0,99989	0,99998	0,99997	0,99997	0,99998	0,99998

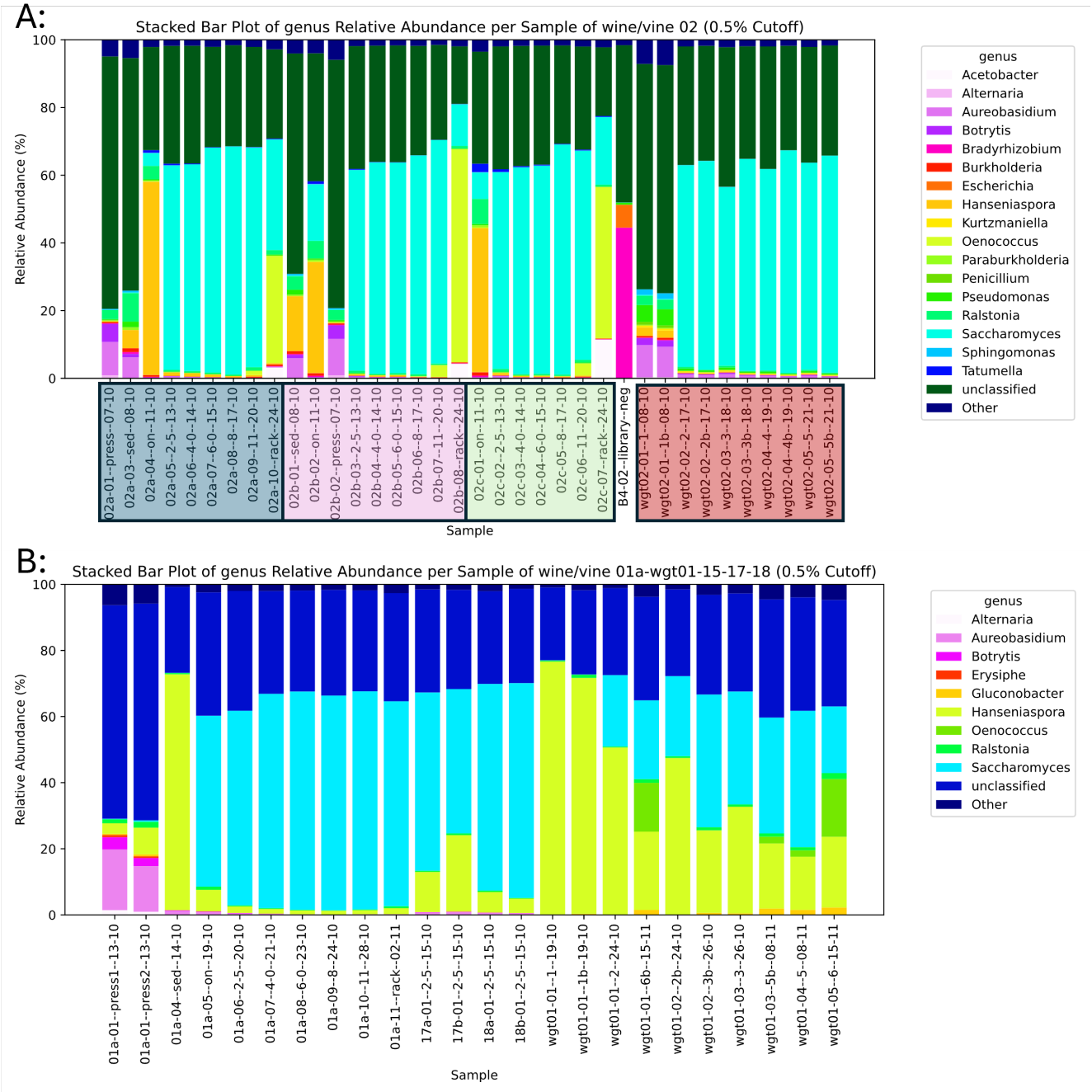
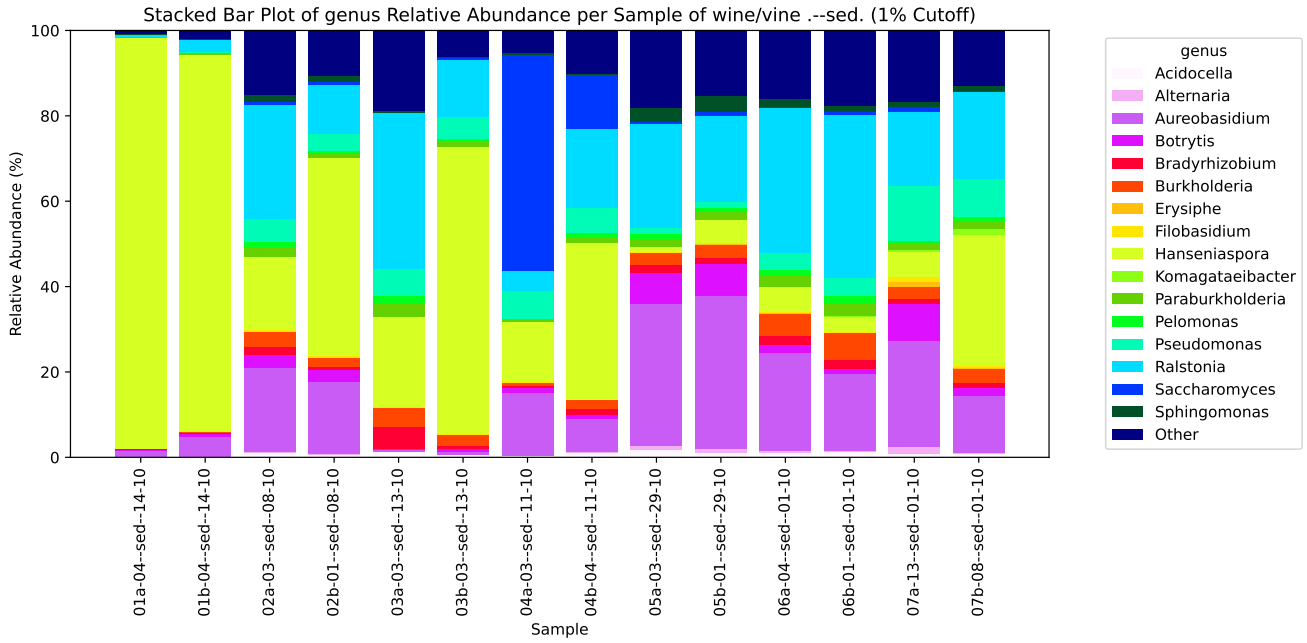


Figure 9. ***Fermentation biological replicates are similar, not same. Winery starts out same as bioreplicates, then changes to be Saccharomyces domination. Hanseniaspora drives fermentation in winery 1, Saccharomyces in sterile 01.***

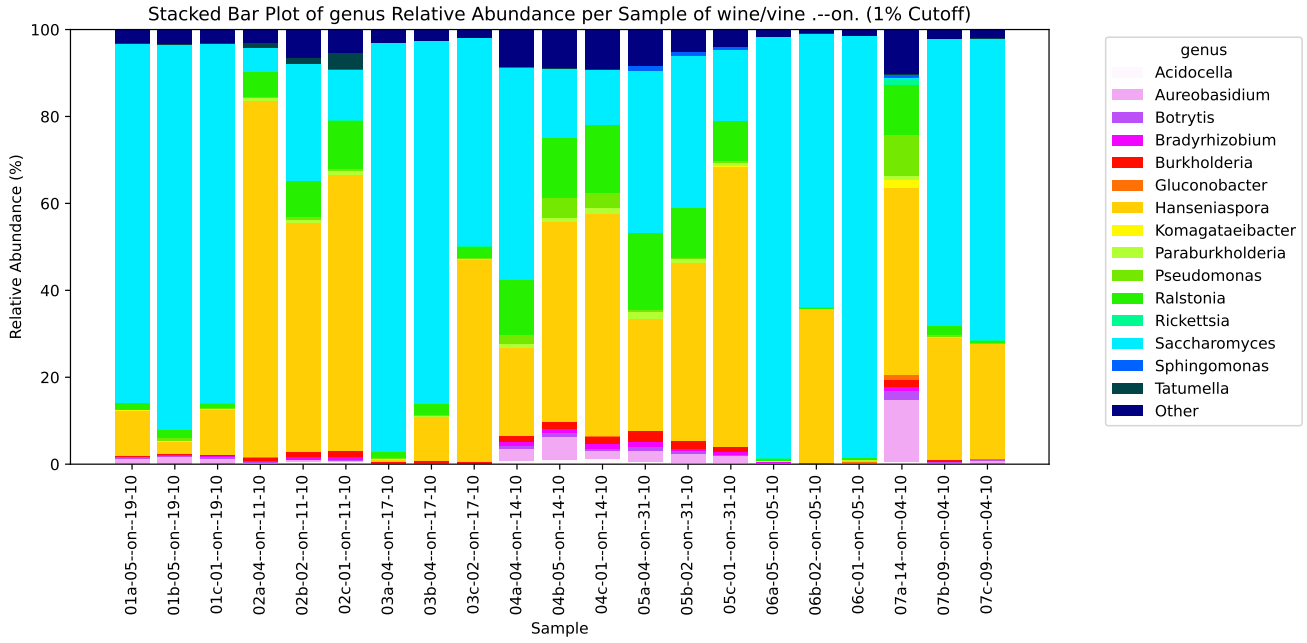
Samples imperfectly sorted due to file naming issues. The order should be: Press, Sedimantation (sed), Onset (on), 2.5, 4, 6, 8, 11, rack. The numbers indicate at which alcohol percentage the sample was taken. The names themselves are written as: Series-Timepoint in series - - sample fermentation stage - - date of measurement, day-month in 2016. Less than 0.5 percent abundance in all of the samples and genus is placed under 'other'.

(**A**) Vineyard and winery 2. Biological replicates separate transparent boxes to inprove clarity of different fermentation. 17 and 18 only have a sample taken at 2.5% alcohol.

(**B**) Vineyard bioreplicate a and winery 1 with inoculation 17 and 18 at 2.5% alcohol. Sterile fermentation has *Aureobasidium* and *Botrytis* presence in press, with *Hanseniaspora* dominance in sedimentation. *Saccaromyces* gains dominance over time in alcoholic fermentation. In winery fermentation, *Hanseniaspora* very present.



(a) Figure A: Fermentation Sedimentation



(b) Figure B: Fermentation Onset

Figure 10. *Hanseniaspora* often has high presence on fermentation onset and about half of the times at sedimentation. Unclassified reads have been removed, leaving only the classified microbes. Only showing fermentations with samples taken at known fermentation onset.
Samples are not perfectly sorted due to file naming issues as the samples were programatically sorted. The order of fermentation stages should be ordered as: Press, Sedimantation (sed), Onset (on), 2.5, 4, 6, 8, 11, rack. The numbers indicate at which alcohol percentage the sample was taken. The names themselves are written as: Series-Timepoint in series - - sample fermentation stage - - date of measurement, day-month in 2016. Less than 1 percent abundance in all of the samples and genus is placed under 'other'. Unclassified is hidden.

8 Exclusion of inconsistent data to mitigate Bias in multi- omics analysis

To allow for multiple omics integration, it is required to handle bias- causing factors. Some deliberations had to be made before the complete object could be created. For the metagenomics dataset, the onset and final measurement were taken as X- datasets. Wineries did not have a written down onset, thus the first measurement was used. Because of the unclear onset analysis the winery data was not used as the first measurement in winery data is not uniform. Inclusion would introduce a form of temporal bias. Some of the first measurements were taken at onset, others at press or sedimentation. Therefore the winery data is best not be used for the correlation research as is. A downside of only using onset and final fermentation is not using a lot of the present data and information. The upside is that we are comparing the same phase of fermentation. With more work the different winery fermentations could potentially be linked to a specific fermentation checkpoint, allowing its use for future work.

To increase understanding about the fermentation onset data and possible influences of bias, the sparseness was calculated. The sparseness explains that fermentation onset data using all data, including the winery data, has a sparseness of 43 percent. This means 43 percent of all cells in the fermentation onset data has a value of 0 (supplementary Figure 20). This could potentially pose an issue for analysis by introducing biases, such as unreliable effect sizes or yield unexpectedly high or low correlation coefficients. A possible way of alleviating the biases is by increasing the minimum number of reads required before taking the correlation into consideration.

Other options could have been chosen instead of only picking the fermentation onset and final for the sterile fermentations and start and end for the time- series data. An effort could have been made to include the winery fermentations by including all fermentation data we do have for both sterile and winery throughout the primary fermentation, which is present for both sterile and winery fermentations. However, due to lack of time this was not possible and due to the winery data not being clearly labeled unlike the sterile fermentation data is.

To summarize, deliberations had to be made to merge all omics data together. To reduce bias as much as possible, only the sterile fermentation onset and finish were included into the omics data. These data had information on which checkpoint the sample was taken. Future work should focus on including more winery data from the metagenomics and time- series data.

9 Integration of multi- omics data reveals difference in flavour with only microbiome being different

To gain insight in how the different -omics integrate and group together next to the sensory data results, the multiple- omics datasets have been plotted next to each other. The integration of the sparse partial least squares analysis contains the time- series start and final, Major, Minor and Thiol metabolites and

microbes onset and final, plotted out in Figure 11. The plot that the X- multi- omics integration is not a one to one comparison to the final flavours. Several groups can still be made out in the X- block which can be found in the Y- block, such as the 05 group with can still be made out on the left side of both X and Y, and the 04 group in the bottom of both blocks.

For elucidation of what drives the overlapping of the groups, a different method allowing separate blocks to be plotted using the same multi- omics data integration, shown in Figure 12. Utilizing this method we can see the separate effects of the datasets, such as the microbes start looking like a funnel, going from wide to narrow and block microbes finish having a cluster on the left side, with numerous other groups spread out.

To compare the prior distinct sour- mouthfeel vineyard 05 group as found in Figure 1, a comparison was made at the different blocks in Figure 12. In which, we can see a the 05 bioreplicates group together in the fermentation start and finish, minor and major metabolites and the microbes start, with 05 being very distinct at fermentation start. It can also be seen in fermentation start that 01 and 06 group close together, with final block Y grouping close together. The similar starting parameters of the grapes, and similar results can be verified using Figure 4, showing the only significant difference in sensory characteristics between 01 and 06 being colour. It can also be seen that microbes start and finish are similar for 01 and 06.

There was also an interesting outlier in the shape of 07a, which can be seen in Figure 1 and Figure 12 block Y. When comparing the different parameters, we can see that 07a has a similar fermentation start and finish, similar minor, major but a difference in thiols. However, the biggest difference is in the microbes start, where 07a is placed far away from replicates 07b and 07c, yet, the final composition in microbes is similar for all three bioreplicates. If we look closer at supplementary Figure 16 C, specifically at the vineyard 07 replicates, in which we see replicate 07a has a diverse component of microbes as compared to 07b and 07c. Bioreplicates 07a has a larger component 'other', and a sizable percentage of the microbes *Pseudomonas*, reported to be related to pyrazine compounds, which in turn are linked to causing a strong odour [48]. Also *Ralstonia*, without any strong literary flavour connections, has been referred to its presence being linked to a significantly changed fermentation environment by inhibiting yeast growth and acid- producing bacteria. *Botrytis* is linked to honey, apricot and mushroom flavours [43] and *Auerobasidium* present, reported to be fructophylic and thus being able to reduce sweetness by consuming fructose while producing polysaccharides, polymers of malic acid and numerous enzymes [49]. When using the same grapes with a different microbial composition shows a different result in final flavour.

In summary, integration of multiple metagenomics, metabolomics, time- series data and sensory panel data allows for finding patterns during the winemaking process. When comparing the integrated data, one- to- one mapping is not straightforward. Block PLS allows for showing various effects within the data, showing separation on the vineyard 05 group and examining outlier 07a, finding only a diverse microbiome as difference compared to its biological replicates, leading to a different flavour outcome.

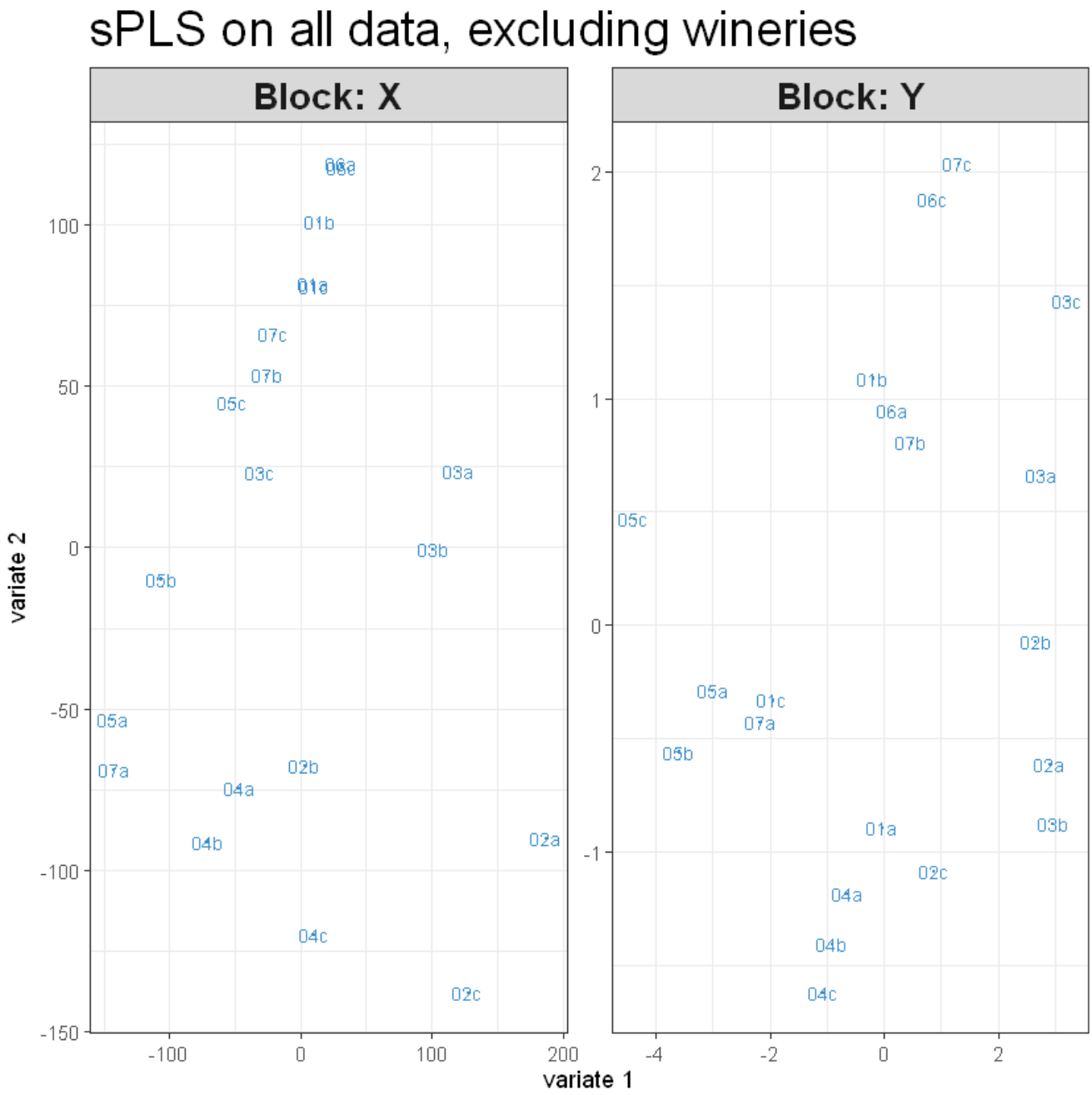


Figure 11. ***Sparse Partial Least Squares on all integrated -omics data show incomplete overlap between all X-omics integrations with the Y- flavours.***

Sparse Partial Least Squares on 'regression' mode with integration of FTIR start, FTIR final, Minor metabolites, Major metabolites, Thiol metabolites, Metagenomics onset and Metagenomics final, compared to Y- sensory profile combination. Seen is not a clear separation when integrating all factors between the different biological replicates, as visible in flavour (Y). When looking at the X- block, the spread of the different samples overlap, causing unclear separation

(**X**) Integrated and scaled time- series fermentation start and end, metagenomics fermentation onset and final, metabolomics major, minor and thiol data. (**Y**) Scaled sensory panel data, causing it to look different to Figure 1.

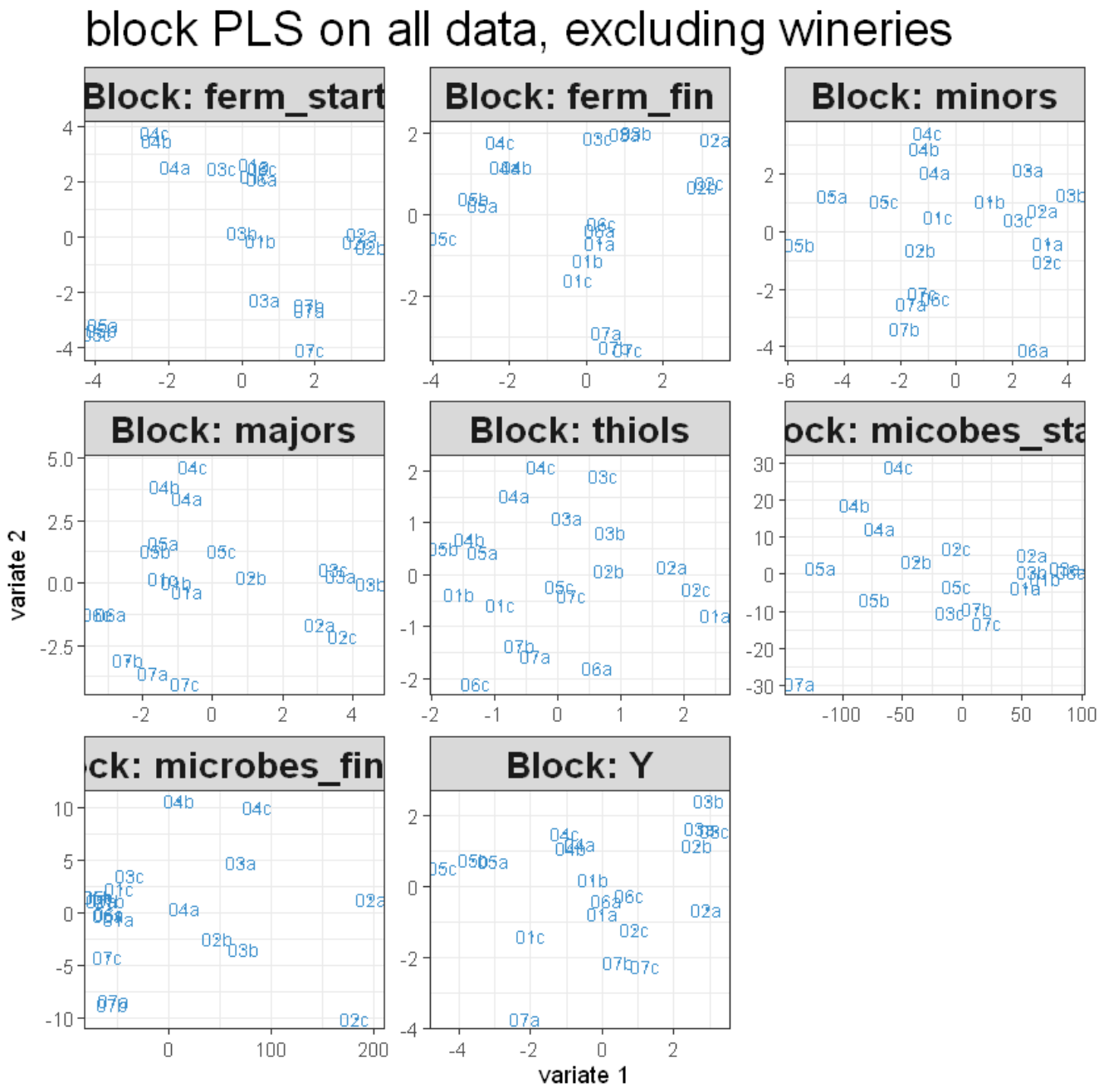


Figure 12. ***Blocked Partial Least Squares on all integrated -omics data show which -omics group as compared to Y- flavours and which -omics perform worse.*** (**a: Explanation**) Blocked Partial Least Squares on 'regresson' mode with integration of FTIR start, FTIR final, Minor metabolites, Major metabolites, Thiol metabolites, Metagenomics onset and Metagenomics final, compared to Y- sensory profile combination. (**b: Visible**) Some -omics such as ferm-start and ferm-fin group the biological replicates close together. The microbes-start and microbes-fin are more spread apart.

10 Results of correlations give insight into data yet warrant caution during interpretation

To make clear connections between individual factors and flavour, correlations were made. All factors were correlated to each flavour, capturing the correlation, P- value and False Discovery Rate (FDR) corrected P-values. As a total of just over 252.000 correlations were calculated, the FDR correction is strict and will only show extremely significant correlations. This results in a massive correlation matrix with 8 FDR significant with P < 0,05 and 47.000 significant P- values, visualised in supplementary Table 8. However, as the multiple testing correction classes nearly all correlations as non- significant, and connections are left out such as the final level of malic acid having a positive link to the experienced level of sour, as known from literature [50]. Yet, malic acid was not significant with FDR correction. Hence, the FDR was used in an advisory role, as the correlations still give an useful insight into the data.

To alleviate bias, less intuitive yet important columns to understanding the data were added. The columns are 'sum_reads' and 'sum_factor' columns These columns are primarily important if the dataset origin is microbes start or finish. The 'sum_reads' column describes how many reads were assigned to that genus and 'sum_factor' takes the sum of the entire column used for correlation. As for example, the organism *Tenebriocola* was found, shown in supplementary Table 9. This organism was investigated due to its correlating significance at fermentation start for mouthfeel, exotic and fruity odour and taste. However, it does have a very low sum reads, warning for a possible biased correlation. *Tenebriocola* is in fact a mealworm beetle with no influence on fermentation. As such, looking at the Figure 20 it could be recommended to filter out low- abundance organisms to guard more strongly against biased results.

Concluding, the multiple testing correction method used was strict and filters out many significant correlations and was therefore judged to be not completely reliable. In place, findings should be considered with due caution. The number of reads utilized to make correlations should be taken into account when assessing reliability of the results.

11 Correlation verification in line with other studies

To verify the calculated Spearman correlations, the results were referred to literature, starting with highly present microbe *Aureobasidium*, as seen in Figure 10. Literature research indicates *Aureobasidium* activity be antagonistic to yeast [49] and antimicrobial in influence [51] and linked to increased mouthfeel of wine [52]. This could be linked to a less intense colour of the wine, which can be connected to less microbial activity [29]. The correlations of *Aureobasidium* in relation to wine sensory properties with P- values < 0,05 are shown in supplementary Table 12. In which we negative correlation to fruity odour, exotic, colour and apple- peach at fermentation onset with low P- values. There are positive correlations for bitter to mouthfeel and sour at fermentation start. The negative colour and positive sour - mouthfeel experience is

in line with the literature stating antagonistic and antimicrobial influences.

To investigate more highly present microbes, known pathogen *Botrytis*, was investigated with all of the Spearman correlations plotted in supplementary Table 10. Literature reports an increase in aromatic intensities for peachy, fruity, floral and toasty notes with an increased oxidization leading to browning of the wine. [53, 54, 55]. The findings of this report indicate an increased *Botrytis* presence causes an increased perception of smokey and green flavours, while decreasing the floral, fruity and exotic perceptions. However, the references are examples of advanced *Botrytis* infection, referring to the infection as noble rot. It is unknown based on the data if *Botrytis* infection is advanced, however it is likely that the infection is milder as compared to literature. There still is a positive link to the smokey flavour, which is related to the toasty note referred in literature, suggesting that lower concentrations of *Botrytis* presence increases green and smokey flavours.

To summarize, the found connections broadly agree with literature. Known organism *Aureobasidium* and *Botrytis* were compared with discovered effects against literature, with a complete overlap in sensory experience *Aureobasidium* compared to literature and differences in the case of *Botrytis*, possibly caused by severity of infection.

12 Comparing multiple influences of multiple organisms to literature finds

Analysis was performed on several organisms based on interest or high presence with varying levels of research with their relation to flavour to get an indication of how the presence influences final wine flavour. The table with connections is shown in supplementary Table 11.

Investigating the *Tatumella* correlations due to its abundance in the start of fermentation, it shows a strong positive correlation to sensory characteristics sweet, bitter, exotic and a negative correlation to sour and honey. Literature links *Tatumella* to have a negative correlation to sour [56] and is positively linked to metabolites causing sweet, fruity, exotic experience [33], which is in line with our findings.

Virusses were also found in the samples, marked with _10239 as per their NCBI number with an increased presence and significance at the end of fermentation. No direct comparisons were made linking a virus to flavour. However, the effects of virusses are only indirect as they only infect a host and do not create their own metabolites. In either case, the virus group is found to be increasing sweet and smokey flavours, while reducing honey and sour flavours. There has been limited studies to viruses in fermentation processes and their influences to the fermentation process, which have primarily been focussed on biologic control [57], but not specifically connecting the influence of viruses to final fermentation flavour. As such, this information is to be considered with a critical perspective.

The main fermentative yeasts were also investigated, as it is interesting to see the effects of some of the more common yeasts and how they align to literature. Some of the more common yeasts found

are *Hanseniaspora* and *Saccharomyces*. Care must be taken when investigating *Hanseniaspora* as the high- *Hanseniaspora* presence was not found in any of the sterile fermentations, potentially leading to incomplete or biased results. Yet still there was some effect found in the form of increasing the experienced green and exotic flavour. Using correlations, differences found in winery 1, having where it could be linked that high *Hanseniaspora* increases floral and honey flavour [47] were not found. Another comparison could be made with the primary fermentative yeast *Saccharomyces*, being linked to modulating mouthfeel and sour, changing the content of polysaccharides within the wine [8, 58], while reducing the fruity experience, body and green due to outcompeting other microbes [59], being in line with literature.

Another interesting group is the *Burkholderia* genus, being present is relatively large quantities as seen in supplementary Figure 16 throughout the fermentations and within the data. The *Burkholderia* group is split up in to *Burkholderia* and *Paraburkholderia*, both giving a similar result in the correlation experiment. Burkholderia and paraburkholderia influence fruity odour positively, green and lemon. *Burkholderia* also influences sour positively extra, but not as significantly as the or strongly correlated as the prior correlations. Both groups negatively influence colour, microbiological and exotic flavours, with *Paraburkholderia* influencing mouthfeel less significant and less strongly correlated. There is little to no direct relation on how the two *Burkholderia* relate to flavours. Some indirect evidence exists, linking the two *Burkholderia* with key volatile metabolites, in turn linked to flavours [60, 61]. Interestingly enough, *Paraburkholderia* was found to have antifungal- antimicrobial properties in at least some strains[62, 63], where this project reveals an antagonistic effect to microbiological - colour sensory experience, which is in line with antimicrobial activity. There were no papers talking directly about *Burkholderia* reducing an exotic sensory experience, nor about metabolites being consumed. Perhaps *Burkholderia* is antagonistic to microbes creating the metabolites, or creates enzymes beraking down metabolites causing exotic flavours.

In a prior section it was found that *Ralstonia* has little to no connections to flavour in winemaking from literature. *Ralstonia* has been linked to be detrimental to the fermentation process and with the recommendation that *Ralstonia* should be inhibited, linking the bacteria to a significantly changed fermentation environment by inhibiting yeast growth and acid- producing bacteria [64]. Therefore it is interesting to investigate the correlations with *Ralstonia*. This project reveals *Ralstonia* to have a positive correlation to sensory characteristics to fruity odour at the start of fermentation and sour with green at the end of fermentation. Negative relations are to colour, exotic, microbiological, mouthfeel and smokey. However, as the correlations go, sour has P- value of 0,044, which is relatively high and should be considered with due caution. However, the trend seems to be in line with the literature. Ralstonia reduces exotic and smokey flavours, possibly by blocking growth of microbes producing the metabolites causing those sensory experiences.

To sum up, the resulting correlations give indications how microbes influence final wine flavour. Common microbes present and researched in wine fermentation have been linked to literature, where this work is in agreement to literature, giving the correlations credibility. Therefore, new links can be found,

such as *Burkholderia* and *Ralstonia*, both highly present, yet not fully investigated organism in relation to fermentation, other than mentioning its presence and indications of influences. These microbes have been linked to final flavours. Also, novel connections with virusses were found, linking the influence of virusses to an increased sensory experience for sweetness and smokey, while decreasing the honey-sour experience. Due to the numerous microbes present influencing final flavour, it can be said that the microbiome influences final flavour substantionally and the resulting correlations can be considered informative.

13 Gaps and Follow- up work

This project mainly focused on its multi- omics integration utilizing classical correlation analysis. As the project is a regression problem, where we create our Y- values from sensory panel analysis and attempt to uncover the mechanisms behind the sensory characteristics. A strong alternative method to using correlations is to utilize supervised machine learning methods. Some of those methods include but are not limited to using (multiple) linear regression, random forests or support vector machines. Due to time constraints it was not possible to perform machine learning. Some more shortcomings are in the form of only utilizing the start and end timepoint of fermentation. The start and end were chosen as they represent a beginning and final grape values. Yet, beginning and end do not account for the numerous changes happening throughout the fermentation. With more data integration, it could be possible to more accurately link effects throughout time in the fermentation. The difficulties lie in the different days samples were taken. The sterile fermentation data is a lot more streamlined with a lot more samples taken, though the measurement time since start of fermentation vary. Winery samples were taken at specific checkpoints, so perhaps a better way would be to integrate more data by utilizing alcohol percentage.

The same lack of data counts for the metagenome analysis: only the onset and and rack were used as they represent the start and finished fermentation. Winery fermentations did not have clearly defined phases, due to lack of labels. When investigating the different starting points of the winery fermentations, visible in supplementary Figure 17d, it seems to follow patterns as seen for fermentation onset, press and sedimentation. Without clear labeling, it is nigh impossible to differentiate and programatically process this data. Therefore, the best method is likely to integrate the winery data is to label the winery fermentations. This can be followed up by integrating all the different checkpoints throughout the fermentation, allowing links between metagenomics and time- series dataset.

The start of a metagenome assembly was performed, but not analyzed due to time constraints. More metagenome- based methods could be utilized, such as enrichment analysis to find out which biological pathways are more and less present. Or a functional analysis, to find out which genes are more or less present. As many unknown reads reads were unclassified by the reference- based analysis, a strong argument could be made to rapidly take advantage of assembly- based methods to shed light on where the

unclassified reads belong instead of only utilizing reference- based methods.

Correlation analysis is convenient, but many of the statistically significant hits could have been subject to low power. As such, the results should be scrutinized to check if the correlations have sufficient statistical power before reaching conclusions, especially for the microbes. Low numbers of reads assigned to an organism could quickly reach statistical significance with low power due to underlying biases, as demonstrated with one example prior in this report. Therefore, other integrative methods could be used. Some methods are more readily usable than others, such as regularized Canonical Correlation Analysis (rCCA) for the mixOmics [65] package already used [66].

Any follow- up project could continue exactly where this project left off to build upon the results of this project as all code utilized is readily available. A follow up could lead to stronger results due to a more varied approach. A pathway enrichment on the metagenome data or gene set enrichment could be a potential treasure trove in increasing the completeness of the information on this project.

Methods

Experimental design

For this project, 4 different types of datasets were generated using grapes originating from 7 vineyards. The grapes after being picked and transported in autoclaved plastic bags they fermented in a sterile environment at Dienstleistungszentrum Ländlicher Raum Rheinpfalz, Neustadt an der Weinstraße, Germany and in industry-like conditions in 6 different wineries. Extra experiments were performed by utilizing commercial yeast inoculation and adding sulphite during fermentation in the 2016 period. These data are: 1) a metagenomics dataset where samples were taken at the onset of different alcohol percentages, namely at 2.5%, 4%, 6%, 8% and 11%. The samples were sequenced using BGISeq paired end sequencing with 5 - 233 million reads per sample (average of 61 million reads) for a total of 292 samples, as some extra experiments were performed. 2) The targeted metabolomics dataset consists of 29 major compounds, 38 minor compounds, 6 thiols and general statistics about the samples, such as sugar content and acidity on the final wine bottles. 3) Sensory panel information consists of 17 sensory characteristics of 3 biological replicates with 2 judge replicates, tasted on different dates, for 20 judges, graded with a number between 0 and 10. 4) A time- series dataset was generated utilizing Fourier-transform infrared spectroscopy (FTIR), following the fermentation of the samples over time at set checkpoints, tracking numerous biochemical parameters including but not limited to the density of the liquid, total sugar, glucose and fructose. A brief overview of the data is given in Table 4 with more details on each on corresponding method sections.

Table 4. Raw data provided for use in project.

Datatype	Number of files	General notes
Major Compounds	1	29 major components, in milligram. 36 rows.
Minor Components	1	38 minor components, in microgram. 36 rows.
Thiols	1	6 thiols, in nanogram. 36 rows.
Fourier Transform Infrared spectroscopy file	1	Information about sampling, time, sugars in gram/litre, alcohol %, acidity, pH. 33 columns, 532 rows.
Sensory Results	1	17 sensory values, ranging between 0 - 10. 20 judges, 2 - 3 replicates. 21 columns, 1101 rows.
Shotgun sequencing results	292	Biological replicates(abc) and 2 technical replicates per sample. Between 5 - 233 million reads per sample.
Mapping file	1	Information on how sequencing relates to the experiments.

Processing, statistics and visualization

For this project R [67] v4.5.0 were used. For R, the packages Compositions [68] v2.0-8, Corrgram [69] v1.14, Reshape[70] v1.4.4, GGPlot [71] v3.5.2, Tidyverse [72] v2.0.0, Tidyr [73] v1.3.1, Viridis [74] v0.6.5, Cluster [75] v2.1.8.1, MixOmics [65] v6.32.0, corrplot [76] v0.95, GGCorrplot [77] v0.1.4.1 and Proxy [78] v0.4-27 were used for and during analysis. The R analysis was performed in ipynb files, saved on Gitlab.

Additional analysis was performed with Python [79] v3.12.9. The packages used are Pandas [80] v2.2, ScikitBio [81] v0.6.3, MatPlotLib [82] v3.10.1, Seaborn [83] v0.13.2 and Numpy [84] v2.2.5.

Finally, Snakemake [85] v8.29.0 was used for creation of a metagenome read processing pipeline. The pipeline utilizes FastP [86] v0.24.0, Kaiju [87] v1.10.1, BWA [88] v2.2.1, Krona [89] v2.8.1, taxonkit [90] v0.19.0 and Samtools [91] v1.21.

A metagenome assembly utilizing Megahit [92] v1.2.9 was also performed, but not further analyzed.

Sensory panel analysis

Sensory analysis was conducted using a dataset comprising 17 sensory attributes across 1101 samples, evaluated by 20 judges. These include but are not limited to the sterile fermentations and the winery fermentations with their biological replicates. The sensory attributes are "bitter", "sweet", "smoky", "microbiological", "body", "oxidized apple", "exotic", "color", "green", "fruity taste", "fruity odor", "apple-peach", "mouthfeel", "floral", "lemon", "honey", "sour".

Some extra experiments took place and are part of the dataset. These are inoculations study, must fermented in the winery 7 (wgt7_t209) was inoculated with commercial yeast 1 (SIHAFERM™ Element, Eaton) and must from vineyard 1 (wb1) was inoculated with two commercial yeast products in fermentation duplicates, yeast 1 (17a,b) (SIHAFERM™ Element, Eaton) and yeast 2 (18a,b)(Fermivin® 4F9, Oenobrands). Furthermore, experiment 15 a - c were inocculated with sulphites. [12].

For sensory analysis, the first step the goal was to visualize the data. For this, Principal Component Analysis (PCA) was performed in order to find out if groups would form. Judge taste replicates were merged on average data. To verify the results of the PCA, raw and normalized (clustered) heatmaps were

generated and compared with the PCA.

The judges themselves were also tested to uncover if there are any specific biases present. This was performed by testing the average flavours as tested by the judges and plotting them out on boxplots with outliers visualized and marked. The coefficient of variation was also calculated between replicates tasted on the different days to find hard to judge flavours. The same results were also averaged out to ease interpretability. The judges were also ranked based on their average coefficient of variation. Flavours were correlated using correlograms with Spearman correlations. Significant differences were highlighted.

Finding differences in flavour between different experiments was performed using ANOVAs with Tukey multiple testing correction. Each judge had at least 2 replicates. As there are 20 judges, it means each ANOVA per flavour has at least 40 datapoints with its resulting P- value. Finally, the fold changes were calculated as compared to a baseline and plotted in a grid plot, compared with each other for each experiment using the formula.

$$\text{Fold Change}_{(\text{baseline vs target})} = \frac{\overline{x}_{\text{baseline}} - \overline{x}_{\text{target}}}{\overline{x}_{\text{target}}}$$

The statistically significant differences in flavour as compared to said baseline were marked to ease finding differences in flavour.

FTIR time- series Data Analysis

These is a Fourier transform infrared spectroscopy (FTIR) dataset containing 33 columns with 532 rows. The FTIR dataset measures the "density", "alc_g", "total_sug", "glu", "fru", "total_acid", "pH", "va", "tart", "lactic", "malic", "citric", "glyc", "extract", "NOPA", "NH4" and "alc_percent" as main parameters used during analysis.

Analysis of the time- series dataset was performed on the FTIR dataset. This was started by calculating the days since start of fermentation or first measurement and preparing both separate and merged bioreplicate datasets for analysis. Merging was performed with averages. Figures were created to track the different variables of the FTIR dataset over time. Further analysis was performed specifically on the grapes using a different location.

Effort was made to with plot malic vs lactic acid utilizing a linked scatterplot, linked by days, as a confirmation to the expectation that malic acid converts to lactic acid by lactic acid bacteria. Additional correlations were calculated in a large matrix comparing all the different parameters using Pearson correlation. Rows with NaN values were removed and the same correlation matrix was calculated again to include all data. The different matrix were compared divided and subtracted from each other to elucidate difference between the two matrix as a method to find reliability of both matrices.

Metabolomics data analysis

A metabolomics dataset is available, having 4 files. A file containing 'majors', 'minors' and 'thiols'. All files contain 36 rows and exclude wgt3. The majors contain 29 different metabolites, measures in mg. The minors contain 38 different metabolites, measured in µg. The thiol file contains 6 different thiols without clear name, measured in µg.

Examining metabolomics was performed by merging the minor, major and thiol data and creating a dendogram to investigate closeness of the final bottles. Calculating the agglomerative coefficient on the "average", "single", "complete", and "ward" clustering. A plot was generated utilizing ward clustering. The data was scaled and split up into wineries and sterile replicates. For each metabolite, the mean was calculated in both sterile replicates and winery, grouping all winery metabolites together and all sterile data together. The differences between means were calculated for both all sterile experiments, including inoculations and for all spontaneous fermentations only.

$$\text{Difference in Mean}_{(\text{Sterile vs Winery})} = \bar{x}_{\text{baseline}} - \bar{x}_{\text{target}}$$

Finally, the merged metabolomics were analyzed by generating PCA plots for each separate metabolite group and plotted together.

Metagenomics data analysis

A metagenomics dataset encompassing 294 files of metagenomics samples ranging between 5 - 233 million reads per sample. An overview of all data shown in Table 4.

All metagenomics files were extracted from their source file and renamed based on the provided carboy file to keep a clear connection to what the metagenomics sample represents. Manual Bash terminal commands were used to extract the metagenomics fastq files from the folder/lane name, appending the lane name to the files. A custom Python script was used to change the filenames to their respective sample name.

Metagenome assembly was performed with Megahit [92]. To assemble on a computer with 2TB RAM available, the technical replicates were not used, and for biological replicates only the a bioreplicate was chosen for the metagenome assembly, with the host genome removed. The reads from the metagenome pipeline were used. The options used were min-count 3, a K- list of 7,37,47,57,67,77,87 and -no-mercy -1. The elapsed time was 54 hours, gaining 6217935 contigs, total 5253925362 bp, min 200 bp, max 816966 bp, avg 844 bp, N50 1012 bp, memory used: 2920690062950. Further analysis was not performed for metagenome assembly based methods due to time constraints.

Cleaning up the raw reads was performed by utilizing FastP [86] to trim reads with a PHRED score

<20. In general the Q20 bases are at 97 percent, with Q30 bases at 90 percent, indicating high quality. To further clean up the data, the shotgun reads had been mapped to the *Vitis Vinifera* [41] reference genome recommended by the NCBI.

A metagenomics pipeline utilizing Kaiju [87] was created with a figure showing the complete pipeline at supplementary Figure 13. This pipeline trims the raw reads on PHRED quality below 20 and common adapters. A reference genome is supplied to the algorithm to filter out host reads. The given host genome and raw reads are indexed using BWA [88]. The indexed reads are aligned to the host genome using Samtools [91]. Forward and reverse read are removed if either aligns to host genome. The reads are separated from the metagenomics file and used for further processing. Statistics are generated describing number of host reads per sample using Samtools. The used host genome is 'GCA_030704535.1 ASM3070453v1' *Vitis vinifera* [41] as retrieved from NCBI as the highlighted reference genome.

The remaining reads undergo Kaiju [87] assignment. For the reference- based assignment, the prebuilt Kaiju index nr-euk-2023-05-10 [93] was used. This index contains an amalgamation of Archaea, bacteria, viruses, fungi and microbial eukaryotes. In total containing just over 8 million taxa.

The results of the classification gets turned into a Krona plot in HTML format. The same information gets placed in tsv files and taxonkit [90] is used to turn accession codes into human readable format in combination with using custom python scripts for classifying on specific levels. The genus was the used level for analysis.

Finally, the a custom Python script utilizing the package scikit-bio [81] is used to calculate the Alpha and Beta diversity within and in between samples. The Alpha Diversity calculations are summarized using 'unclassified', 'sum-counts' , 'percent-unclassified', 'ShannonDiversity', 'AceDiversity', 'Chao1Diversity', 'SimpsonEveness', 'GiniDominance' and 'GoodsCoverage'. For the Beta diversity, the statistics were calculated using the Bray-Curtis method while excluding the unclassified output, generating a correlation matrix between samples. Using the Alpha counts and unclassified statistics, boxplots on the different taxonomic levels are generated.

Multi- omics approach

All the omics data was brought together within a R- script utilizing the MixOmics [65] package in a ipynb file. The selected files are average flavours as judged by the judges, FTIR timeseries dataset start and final values, metabolomics, Kaiju microbial composition at start of fermentation (fermentation onset) and end of fermentation (rack). Final values of the wines were calculated on the average of the biological replicates.

The data was scaled, after which PCA plots were generated on the separate datasets. Further PCA plots were generated by adding FTIR start and final values, all the while investigating loadings. Continuing

with addition of the metabolomics dataset by combining the majors, minors and thiol datasets, creating more PCAs and a correlation heatmap of the intermediary object. Merging and analysis the different integrations was performed using Sparse Partial Least Squares (SPLS) and Block Partial Least Squares (Block PLS).

The sparseness was calculated and plotted for the fermentation onset data to see how much percent of the metagenomics data was under a specific number of reads. Finally, the microbe composition at onset and final were added to the multi- omics approach and more PCAs were plotted with corresponding correlation heatmaps. Cutoffs at 0.3 were added to PCA.

With the results of the PCA, euclidean distances were calculated between each datapoint and each flavour type and exported to a .tsv file. Manually investigation occurred, searched through the data for close datapoints to flavours and see the closeness of factors to flavours. Some factors were manually checked in their closeness by correlating the flavour to factor using spearman correlation and plotting the datapoints. Loadings were also investigated for the final PCA.

Continuing, correlations with different factors were calculated to final flavour types. The Gini dominance and Shannon diversity were also plotted out in a boxplot to see how many of the fermentations were highly dominated by one genus. Finally, all correlations were calculated for the different factors from the combined dataset against flavours for only the sterile fermentations. P- values were calculated for all, just over 252.000, combinations and FDR correction was applied on all combinations at once. From these results, the factor, correlation, P-value and adjusted P-value were saved. As extra information, the sum of the factor was also added, mainly for the microbe fermentation as indication how many reads were present throughout all fermentations to account for biases.

Some flavour - factor combinations were more closely analyzed and compared to literature to cross reference findings and possibly find new undocumented links between factor and flavour.

Artificial Intelligence statement

Throughout this project, AI was used to assist in programming snippets of code and as sparring partner, asking questions and manually verifying the answers. AI was further used to assist in literary research, to aid in finding suitable scientific articles in combination with traditional article searching. AI- suggested articles were manually verified for its relevance. AI was not used for writing this report, no AI generated sections are present.

Data statement

The resulting Kaiju pipeline is available on WUR Gitlab, Meint Eybergen. The remaining parts of analysis are provided on WUR Gitlab, Dominic Hildebrand.

Collaboration statement

The metagenomics pipeline was created in cooperation with Meint Eybergen, meint.eybergen@wur.nl.

References

1. Li Liu, Jiajing Wang, Danny Rosenberg, Hao Zhao, György Lengyel, and Dani Nadel. Fermented beverage and food storage in 13,000y-old stone mortars at raqefet cave, israel: Investigating natufian ritual feasting. 21:783–793.

2. Klaus Buchholz and John Collins. The roots—a short history of industrial microbiology and biotechnology. 97(9):3747–3762.

3. Kendall A. Smith. Louis pasteur, the father of immunology? 3:68.

4. Maria Jesus Ortiz, Nuria Barrajón, Milla Alves Baffi, Maria Arévalo-Villena, and Ana Briones. Spontaneous must fermentation: Identification and biotechnological properties of wine yeasts. 50(2):371–377.

5. Laura Canonico, Alice Agarbati, Francesca Comitini, and Maurizio Ciani. Assessment of spontaneous fermentation and non-saccharomyces sequential fermentation in verdicchio wine at winery scale. 8(3):49.

6. Francesca Comitini, Mirko Gobbi, Paola Domizio, Cristina Romani, Livio Lencioni, Ilaria Mannazzu, and Maurizio Ciani. Selected non-saccharomyces wine yeasts in controlled multistarter fermentations with saccharomyces cerevisiae. *Food Microbiology*, 28(5):873–882, 2011.

7. Di Liu, Jean-Luc Legras, Pangzhen Zhang, Deli Chen, and Kate Howell. Diversity and dynamics of fungi during spontaneous fermentations and association with unique aroma profiles in wine. *International Journal of Food Microbiology*, 338:108983, 2021.

8. J.h. Swiegers, E.j. Bartowsky, P.a. Henschke, and I.s. Pretorius. Yeast and bacterial modulation of wine aroma and flavour. 11(2):139–173. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1755-0238.2005.tb00285.x.

9. Maurizio Ciani, Angela Capece, Francesca Comitini, Laura Canonico, Gabriella Siesto, and Patrizia Romano. Yeast interactions in inoculated wine fermentation. 7. Publisher: Frontiers.

10. Delphine Jussier, Amélie Dubé Morneau, and Ramón Mira de Orduña. Effect of simultaneous inoculation with yeast and bacteria on fermentation kinetics and key wine parameters of cool-climate chardonnay. 72(1):221–227.

11. Talat Mehmood, Tariq Masud, Syed Ali Abbass, and Shabana Maqsud. Isolation and identification of wild strains of lactic acid bacteria for yoghurt preparation from indigenous dahi. 8(6):866–871.

12. Kimmo Sirén, Sarah Siu Tze Mak, Chrats Melkonian, Christian Carøe, Jan Hendrik Swiegers, Douwe Molenaar, Ulrich Fischer, and M. Thomas P. Gilbert. Taxonomic and functional characterization of the microbial community during spontaneous in vitro fermentation of riesling must. 10. Publisher: Frontiers.

13. Justin D. Silverman, Rachael J. Bloom, Sharon Jiang, Heather K. Durand, Eric Dallow, Sayan Mukherjee, and Lawrence A. David. Measuring and mitigating PCR bias in microbiota datasets. 17(7):e1009113.

14. Tamara Milivojević, Shirin Nurshan Rahman, Débora Raposo, Michael Siccha, Michal Kucera, and Raphaël Morard. High variability in SSU rDNA gene copy number among planktonic foraminifera revealed by single-cell qPCR. 1(1):63. Publisher: Nature Publishing Group.

15. Christopher Quince, Alan W. Walker, Jared T. Simpson, Nicholas J. Loman, and Nicola Segata. Shotgun metagenomics, from sampling to analysis. 35(9):833–844. Publisher: Nature Publishing Group.

16. J. T. Lennon, M. E. Muscarella, S. A. Placella, and B. K. Lehmkuhl. How, when, and where relic DNA affects microbial diversity. 9(3):10.1128/mbio.00637–18. Publisher: American Society for Microbiology.

17. Augusto Dulanto Chiang and John P Dekker. From the pipeline to the bedside: Advances and challenges in clinical metagenomics. 221:S331–S340.

18. Qi Peng, Jiaxin Huang, Shanshan Li, Zeyu Chen, Qing Zhu, Hexiang Yuan, Jiachen Li, Beatrice Bassilekin Massou, and Guangfa Xie. Dynamics of microbial communities and metabolites during the fermentation of ningxia goji berry wine: An integrated metagenomics and metabolomics approach. 201:115609.

19. Zihua Liang, Shiyun Chen, Hao Wang, Qi Wu, Weiling Guo, Li Ni, and Xucong Lv. Metagenomic and metabolomic profiling reveals the differences of flavor quality between hongqu rice wines fermented with gutian qu and wuyi qu. 13(19):3114. Number: 19 Publisher: Multidisciplinary Digital Publishing Institute.

20. Guanmian Wei, Ziyi Zhang, Feiran Zhao, Yaxin Sang, Joe M. Regenstein, and Peng Zhou. Characteristic aroma compounds during the fermentation of chinese steamed bread fermented with different starters. 457:140151.

21. Se Hee Lee, Tae Woong Whon, Seong Woon Roh, and Che Ok Jeon. Unraveling microbial fermentation features in kimchi: from classical to meta-omics approaches. 104(18):7731–7744.

22. Yu Mu, Wei Su, Yingchun Mu, and Li Jiang. Combined application of high-throughput sequencing and metabolomics reveals metabolically active microorganisms during panxian ham processing. 10. Publisher: Frontiers.

23. Diana Lynne Hawkins, Jess Ryder, Soon A Lee, Katie Parish-Virtue, Bruno Fedrizzi, Matthew R Goddard, and Sarah J Knight. Mixed yeast communities contribute to regionally distinct wine attributes. 23:foad005.

24. Vicente Ferreira and Ricardo Lopez. The actual and potential aroma of winemaking grapes. 9(12):818.

25. Neil P. Jolly, Cristian Varela, and Isak S. Pretorius. Not your ordinary yeast: non-saccharomyces yeasts in wine production uncovered. 14(2):215–237.

26. James Willwerth, Andrew Reynolds, and Isabelle Lesschaeve. Sensory analysis of ontario riesling wines from various water status zones. 52(2):145–171. Number: 2.

27. Weingeschmacksrichtungen.

28. Judit Diószegi, Erand Llanaj, and Róza Ádány. Genetic background of taste perception, taste preferences, and its nutritional implications: A systematic review. 10:1272.

29. Rosanna Tofalo, Giovanna Suzzi, and Giorgia Perpetuini. Discovering the influence of microorganisms on wine color. 12:790935.

30. SIHAFERM element.

31. Fermivin 4f9.

32. José Pedro Santos, Jesús Lozano, Manuel Aleixandre, Teresa Arroyo, Juan Mariano Cabellos, Mar Gil, and Maria del Carmen Horrillo. Threshold detection of aromatic compounds in wine with an electronic nose and a human sensory panel. 80(5):1899–1906.

33. Qinqin Liu, Nan Hao, Lan Mi, Shuai Peng, Akumawah Kyen Marie-Colette, Xuefang Zhao, and Jing Wang. From microbial communities to aroma profiles: A comparative study of spontaneous fermentation in merlot and cabernet sauvignon wines. 26:102317.

34. Jordi Tronchoni, Amparo Gamero, Francisco Noé Arroyo-López, Eladio Barrio, and Amparo Querol. Differences in the glucose and fructose consumption profiles in diverse *Saccharomyces* wine species and their hybrids during grape juice fermentation. 134(3):237–243.

35. N.J. Berthels, R.R. Cordero Otero, F.F. Bauer, J.M. Thevelein, and I.S. Pretorius. Discrepancy in glucose and fructose utilisation during fermentation by saccharomyces cerevisiae wine yeast strains. 4(7):683–689.

36. Torben Bo Toldam-Andersen, Shujuan Zhang, Jing Liu, Wender L. P. Bredie, and Mikael Agerlin Petersen. Sulfite management during vinification and impact on the flavor of solaris wine. 10(4):210. Number: 4 Publisher: Multidisciplinary Digital Publishing Institute.

37. Sydney Christian Morgan. THE EFFECT OF SULFUR DIOXIDE ADDITION AND ALTERNATIVE FERMENTATION TECHNIQUES ON THE MICROBIAL COMMUNITIES AND SENSORY PROFILES OF WINE.

38. N. Yéramian, C. Chaya, and J. A. Suárez Lepe. L-(-)-malic acid production by saccharomyces spp. during the alcoholic fermentation of wine (1). 55(3):912–919.

39. Charlotte Vion, Maitena Muro, Margaux Bernard, Bruce Richard, Fautre Valentine, Nadine Yeramian, Isabelle Masneuf-Pomarède, Sophie Tempère, and Philippe Marullo. New malic acid producer strains of *Saccharomyces cerevisiae* for preserving wine acidity during alcoholic fermentation. 112:104209.

40. Carla Virdis, Krista Sumby, Eveline Bartowsky, and Vladimir Jiranek. Lactic acid bacteria in wine: Technological advances and evaluation of their functional role. 11:612118.

41. Vitis vinifera genome assembly ASM3070453v1.

42. Reid G. Griggs, Lena Flörl, Michael Swadener, Rodrigo Hernández-Velázquez, David A. Mills, and Nicholas A. Bokulich. A tale of two vineyards: parsing site-specific differences in bacterial and fungal communities of wine grapes from proximal vineyards, and their changes during processing in a single winery. Pages: 2025.03.05.641676 Section: New Results.

43. Cátia Pinto, Diogo Pinho, Remy Cardoso, Valéria Custódio, Joana Fernandes, Susana Sousa, Miguel Pinheiro, Conceição Egas, and Ana C. Gomes. Wine fermentation microbiome: a landscape from different portuguese wine appellations. 6. Publisher: Frontiers.

44. Zhong Zhang, Huiqing Wang, Hongchuan Xia, Lijun Sun, Qingchen Zhang, Hui Yang, and Junxiang Zhang. Wine aroma modification by *Hanseniaspora uvarum*: A multiple-step strategy for screening potential mixed starters. 20:100930.

45. Cristobal A. Onetto, Jane McCarthy, and Simon A. Schmidt. A rapid growth rate underpins the dominance of hanseniaspora uvarum in spontaneous grape juice fermentations. Pages: 2024.08.16.608365 Section: New Results.

46. Tess Snyder, James Osborne, and Chris Curtin. Contributions of hanseniaspora species to pinot noir microbial terroir in oregon's willamette valley wine region. 90(9):e00810–24. Publisher: American Society for Microbiology.

47. Elliot Borren and Bin Tian. The important contribution of non-saccharomyces yeasts to the aroma complexity of wine: A review. 10(1):13.

48. Meiqi Wang, Jiarong Wang, Jiawei Chen, Christian Philipp, Xiaoning Zhao, Jie Wang, Yaqiong Liu, and Ran Suo. Effect of commercial yeast starter cultures on cabernet sauvignon wine aroma compounds and microbiota. 11(12):1725.

49. Cristobal A. Onetto, Anthony R. Borneman, and Simon A. Schmidt. Investigating the effects of *Aureobasidium pullulans* on grape juice composition and fermentation. 90:103451.

50. Claire Payan, Anne-Laure Gancel, Michael Jourdes, Monika Christmann, and Pierre-Louis Teissedre. Wine acidification methods: a review. 57(3):113–126.

51. Mahendra Rai, Magdalena Wypij, Avinash P. Ingle, Joanna Trzcińska-Wencel, and Patrycja Golińska. Emerging trends in pullulan-based antimicrobial systems for various applications. 22(24):13596.

52. Despina Bozoudi and Dimitrios Tsaltas. The multiple and versatile roles of aureobasidium pullulans in the vitivinicultural sector. 4(4):85. Number: 4 Publisher: Multidisciplinary Digital Publishing Institute.

53. I. Ky, B. Lorrain, M. Jourdes, G. Pasquier, M. Fermaud, L. Gény, P. Rey, B. Doneche, and P.-L. Teissedre. Assessment of grey mould (botrytis cinerea) impact on phenolic and sensory quality of bordeaux grapes, musts and wines for two consecutive vintages. 18(2):215–226. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1755-0238.2012.00191.x.

54. Angela Lopez Pinar, Doris Rauhut, Ernst Ruehl, and Andrea Buettner. Quantification of the changes in potent wine odorants as induced by bunch rot (botrytis cinerea) and powdery mildew (erysiphe necator). 5. Publisher: Frontiers.

55. Angela Lopez Pinar, Doris Rauhut, Ernst Ruehl, and Andrea Buettner. Effects of *Botrytis cinerea* and *Erysiphe necator* fungi on the aroma character of grape must: A comparative approach. 207:251–260.

56. Alena M. Bubeck, Lena Preiss, Anna Jung, Elisabeth Dörner, Daniel Podlesny, Marija Kulis, Cynthia Maddox, Cesar Arze, Christian Zörb, Nikolaus Merkt, and W. Florian Fricke. Bacterial microbiota diversity and composition in red and white wines correlate with plant-derived DNA contributions and botrytis infection. 10:13828.

57. Bruna L Maske, Dão P De Carvalho Neto, Gabriela B da Silva, Juliano De Dea Lindner, Carlos R Soccol, and Gilberto V de Melo Pereira. Yeast viruses and their implications in fermented foods and beverages. 47:100879.

58. R. Gawel, P. A. Smith, and E. J. Waters. Influence of polysaccharides on the taste and mouthfeel of white wine. 22(3):350–357. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/ajgw.12222.

59. Taylor Reiter, Rachel Montpetit, Shelby Byer, Isadora Frias, Esmeralda Leon, Robert Viano, Michael Mcloughlin, Thomas Halligan, Desmon Hernandez, Ron Runnebaum, and Ben Montpetit. Saccharomyces cerevisiae gene expression during fermentation of pinot noir wines at an industrially relevant scale. 87(11):e00036–21.

60. Wei Wu, Zhuochen Wang, Boyang Xu, Jing Cai, Jianghua Cheng, Dongdong Mu, Xuefeng Wu, and Xingjiang Li. Exploring core microbiota based on characteristic flavor compounds in different fermentation phases of sufu. 27(15):4933.

61. Guimei Chen, Yujie Yuan, Suwen Tang, Ziyi Yang, Qi Wu, Zihua Liang, Shiyun Chen, Wenlong Li, Xucong Lv, and Li Ni. Comparative analysis of microbial communities and volatile flavor components in the brewing of hongqu rice wines fermented with different starters. 7:100628.

62. Gordon Webster, Alex J. Mullins, Aimee S. Bettridge, Cerith Jones, Edward Cunningham-Oakes, Thomas R. Connor, Julian Parkhill, and Eshwar Mahenthiralingam. The genome sequences of three paraburkholderia sp. strains isolated from wood-decay fungi reveal them as novel species with antimicrobial biosynthetic potential. 8(34):10.1128/mra.00778–19. Publisher: American Society for Microbiology.

63. Mariana Ferreira Rabelo Fernandes, Thiago Gonçalves Ribeiro, Janaína Ribeiro Rouws, Luis Henrique de Barros Soares, and Jerri Édson Zilli. Biotechnological potential of bacteria from genera bacillus paraburkholderia and pseudomonas to control seed fungal pathogens. 52(2):705–714.

64. Yao Lu, Xiaolei Guan, Rongbin Li, Jie Wang, Yaqiong Liu, Yanli Ma, Jiawei Lv, Suwen Wang, and Jianlou Mu. Comparative study of microbial communities and volatile profiles during the inoculated and spontaneous fermentation of persimmon wine. 100:49–58.

65. Rohart F, Gautier B, Singh A, and Le Cao K-A. mixomics: An r package for 'omics feature selection and multiple data integration. *PLoS computational biology*, 13(11):e1005752, 2017.

66. Aurelia Morabito, Giulia De Simone, Roberta Pastorelli, Laura Brunelli, and Manuela Ferrario. Algorithms and tools for data-driven omics integration to achieve multilayer biological insights: a narrative review. 23:425.

67. R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria, 2025.

68. K. Gerald van den Boogaart, Raimon Tolosana-Delgado, and Matevz Bren. *compositions: Compositional Data Analysis*, 2024. R package version 2.0-8.

69. Kevin Wright. *corrgram: Plot a Correlogram*, 2021. R package version 1.14.

70. Hadley Wickham. Reshaping data with the reshape package. *Journal of Statistical Software*, 21(12):1–20, 2007.

71. Hadley Wickham. *ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York, 2016.

72. Hadley Wickham, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani. Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686, 2019.

73. Hadley Wickham, Davis Vaughan, and Maximilian Girlich. *tidyr: Tidy Messy Data*, 2024. R package version 1.3.1.

74. Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Antônio Pedro, Sciaini, Marco, Scherer, and Cédric. *viridis(Lite) - Colorblind-Friendly Color Maps for R*, 2024. viridis package version 0.6.5.

75. Martin Maechler, Peter Rousseeuw, Anja Struyf, Mia Hubert, and Kurt Hornik. *cluster: Cluster Analysis Basics and Extensions*, 2025. R package version 2.1.8.1 — For new features, see the 'NEWS' and the 'Changelog' file in the package source).

76. Taiyun Wei and Viliam Simko. *R package 'corrplot': Visualization of a Correlation Matrix*, 2024. (Version 0.95).

77. Alboukadel Kassambara. *ggcorrplot: Visualization of a Correlation Matrix using 'ggplot2'*, 2023. R package version 0.1.4.1.

78. David Meyer and Christian Buchta. *proxy: Distance and Similarity Measures*, 2022. R package version 0.4-27.

79. Guido Van Rossum and Fred L. Drake. *Python 3 Reference Manual*. CreateSpace.

80. The pandas development team. pandas-dev/pandas: Pandas, February 2020.

81. http://scikit-bio.org The scikit-bio development team. scikit-bio: A bioinformatics library for data scientists, students, and developers, 2020.

82. J. D. Hunter. Matplotlib: A 2d graphics environment. 9(3):90–95. Publisher: IEEE COMPUTER SOC.

83. Michael L. Waskom. seaborn: statistical data visualization. 6(60):3021. Publisher: The Open Journal.

84. Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. 585(7825):357–362. Publisher: Springer Science and Business Media LLC.

85. Felix Mölder, Kim Philipp Jablonski, Brice Letcher, Michael B. Hall, Christopher H. Tomkins-Tinch, Vanessa Sochat, Jan Forster, Soohyun Lee, Sven O. Twardziok, Alexander Kanitz, Andreas Wilm, Manuel Holtgrewe, Sven Rahmann, Sven Nahnsen, and Johannes Köster. Sustainable data analysis with snakemake.

86. Shifu Chen, Yanqing Zhou, Yaru Chen, and Jia Gu. fastp: an ultra-fast all-in-one FASTQ preprocessor. 34(17):i884–i890.

87. Peter Menzel, Kim Lee Ng, and Anders Krogh. Fast and sensitive taxonomic classification for metagenomics with kaiju. 7(1):11257. Publisher: Nature Publishing Group.

88. Md. Vasimuddin, Sanchit Misra, Heng Li, and Srinivas Aluru. Efficient architecture-aware acceleration of BWA-MEM for multicore systems. In *2019 IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, pages 314–324. ISSN: 1530-2075.

89. Brian D. Ondov, Nicholas H. Bergman, and Adam M. Phillippy. Interactive metagenomic visualization in a web browser. 12:385.

90. Wei Shen and Hong Ren. TaxonKit: A practical and efficient NCBI taxonomy toolkit. 48(9):844–850.

91. Petr Danecek, James K Bonfield, Jennifer Liddle, John Marshall, Valeriu Ohan, Martin O Pollard, Andrew Whitwham, Thomas Keane, Shane A McCarthy, Robert M Davies, and Heng Li. Twelve

years of SAMtools and BCFtools. 10(2). _eprint: https://academic.oup.com/gigascience/article-pdf/10/2/giab008/36332246/giab008.pdf.

92. Dinghua Li, Chi-Man Liu, Ruibang Luo, Kunihiko Sadakane, and Tak-Wah Lam. MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de bruijn graph. 31(10):1674–1676.

93. Kaiju: Fast and sensitive taxonomic classification for metagenomics https://bioinformatics-centre.github.io/kaiju/downloads.html.

Supplementary material

Table 5. Top 5 most different metabolites as shown in Figure 5 D. Scaled results.

	Butyric.acid	Ethyl.acetate	Acetovanillone	c.3.Hexenol	Isoamyl.acetate
01a	-0,411	-0,715	-0,571	-0,862	-0,828
01b	-0,375	-0,300	-0,370	-0,862	-0,658
01c	-0,828	-0,590	-0,853	-0,862	-0,816
02a	0,885	-0,283	-0,196	0,405	0,167
02b	0,995	-0,246	-0,437	0,154	0,536
02c	0,320	-2,555	-0,071	0,439	0,568
03a	0,639	0,938	-0,586	-0,862	1,298
03b	0,697	0,571	-0,667	-0,862	1,204
03c	0,320	0,639	-0,652	0,192	1,013
04a	-0,686	0,698	-0,455	-0,862	-0,221
04b	0,025	0,415	-0,304	-0,862	-0,436
04c	-0,584	0,523	-0,827	0,200	-0,279
05a	-0,373	0,265	-0,644	0,358	0,312
05b	-0,238	0,609	-0,580	0,438	-0,054
05c	-0,064	-0,111	-0,684	0,438	0,301
06a	-0,970	-0,577	0,167	-0,862	-1,213
06c	-1,235	-0,528	-0,068	-0,862	-1,196
07a	-0,341	-0,107	-0,363	-0,862	-0,135
07b	-0,403	-0,416	-0,414	-0,862	-0,475
07c	0,138	0,285	-0,195	-0,862	0,177
15a	-0,626	-0,370	-0,793	-0,862	-0,929
15b	-0,627	-0,979	-0,408	-0,862	-1,031
15c	-1,159	-0,477	-0,482	0,995	-0,775
17a	-1,108	-0,483	-0,686	-0,862	-0,630
17b	-0,976	-0,346	-0,544	-0,862	-0,559
18a	-0,706	-0,777	-1,012	-0,862	-1,287
18b	-0,728	-0,802	-0,701	-0,862	-1,257
wgt1	2,089	3,760	-0,354	2,288	1,062
wgt2	1,511	0,885	0,926	2,529	1,662
wgt4	1,219	0,897	2,253	0,766	0,802
wgt5	3,298	1,542	2,907	1,133	3,241
wgt6F77	-0,494	-0,829	1,714	1,384	-0,415
wgt6F775	-0,142	-0,841	1,600	1,407	-0,411
wgt6T202	-0,911	-0,704	1,766	0,295	-0,832
wgt7F30	0,603	0,315	1,372	1,095	1,049
wgt7T209	1,248	0,694	1,210	0,995	1,043

Table 6. All top 10 most variable bioreplicates of within- sterile fermentation as found by Table 2.

	t.Whiskylactone	Acetaldehyde	MP	Acetic.acid	Decalactone	Ionone	1Hexanol	MOH	cWhiskylactone	Phenylethyl.acetate
01a	0,000	0,139	5,600	210,303	0,000	0,000	1,240	156,300	0,000	59,774
01b	0,000	0,000	0,000	297,994	3,864	0,000	1,173	70,300	9,160	59,320
01c	0,000	0,000	0,000	301,295	3,007	0,000	1,216	77,100	10,791	46,679
02a	0,000	0,000	2,900	152,533	0,000	0,000	1,333	123,200	0,000	82,675
02b	0,000	0,526	3,500	591,882	0,000	0,000	1,326	111,500	7,531	79,291
02c	0,530	0,243	4,400	144,138	0,000	0,000	1,230	118,000	0,000	98,438
03a	0,397	0,150	0,000	367,659	0,000	0,000	1,111	82,000	0,000	79,035
03b	0,000	0,169	0,000	336,820	0,000	0,785	1,198	102,500	0,000	81,592
03c	0,488	0,184	0,000	355,508	0,000	0,000	1,148	37,900	0,000	83,540
04a	1,672	0,138	0,000	441,885	0,000	0,000	1,701	27,300	7,973	41,480
04b	0,000	2,538	0,000	382,585	7,771	0,000	1,752	47,400	9,571	44,156
04c	0,000	0,000	0,000	387,398	1,508	0,000	1,795	30,100	3,275	35,432
05a	0,000	0,000	0,000	239,690	0,000	0,000	1,319	94,500	6,910	162,065
05b	0,000	0,000	0,000	216,383	0,839	0,000	1,321	117,700	5,545	128,249
05c	0,000	0,166	0,000	282,907	0,000	0,000	1,346	128,300	0,000	151,713
06a	0,000	0,146	0,000	315,562	0,000	0,000	1,119	184,500	0,000	30,531
06c	0,000	0,000	0,000	313,091	6,556	0,000	1,139	171,100	8,342	25,246
07a	0,000	0,000	0,000	190,293	0,000	0,000	1,130	154,300	7,986	47,987
07b	9,578	0,000	0,000	327,744	0,000	0,000	1,223	139,700	8,588	45,010
07c	0,000	0,000	0,000	230,884	0,000	0,000	1,013	137,000	4,327	67,027

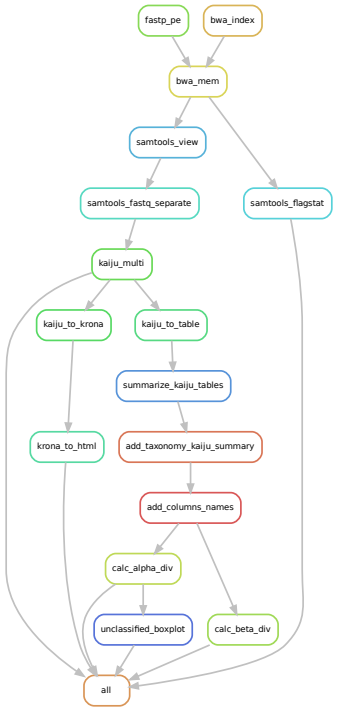


Figure 13. Snakemake- generated dot diagram showing the Kaiju- based metagenome analysis pipeline. The pipeline utilizes FastQ to trim adapters and quality. Kaiju is run for reference based assignment and the results are plotted in a Krona plot and converted to a tsv table, which gets their classification levels added to genus level. Alpha and Beta diversity get calculated, and the level of unclassified genus gets plotted.

	density	alc_g	total_sug	glu	fru	total_acid	pH	va	tart	lactic	malic	citric	glyc	extract	NOPA	NH4	alc_%
density	1	-0,74298	0,730036	0,626525	0,792812	0,378599	-0,4091	-0,18742	0,503411	0,137649	0,227045	0,386294	-0,59634	0,565627	0,210017	0,351867	-0,73365
alc_g	-0,74298	1	-0,98864	-0,96603	-0,98093	-0,40198	0,519852	0,403414	-0,42771	-0,22829	-0,40724	-0,80238	0,945742	-0,77568	-0,49513	-0,55362	0,999783
total_sug	0,730036	-0,98864	1	0,982337	0,989141	0,319743	-0,4419	-0,41283	0,364882	0,281884	0,353821	0,831378	-0,94465	0,747124	0,521371	0,566589	-0,98866
glu	0,626525	-0,96603	0,982337	1	0,944332	0,270644	-0,40994	-0,45068	0,284431	0,316156	0,358508	0,908058	-0,96466	0,7362	0,581047	0,621769	-0,96977
fru	0,792812	-0,98093	0,989141	0,944332	1	0,341124	-0,44856	-0,37389	0,413183	0,251368	0,334941	0,750936	-0,90524	0,732667	0,462903	0,510025	-0,97802
total_acid	0,378599	-0,40198	0,319743	0,270644	0,341124	1	-0,82405	0,063196	0,876294	-0,21379	0,654455	0,132927	-0,32745	0,664446	-0,13778	0,179805	-0,39832
pH	-0,4091	0,519852	-0,4419	-0,40994	-0,44856	-0,82405	1	0,3906	-0,77823	0,177861	-0,40705	-0,35288	0,47701	-0,45428	0,250382	-0,2	0,519319
va	-0,18742	0,403414	-0,41283	-0,45068	-0,37389	0,063196	0,3906	1	-0,03463	0,013902	0,106449	-0,55426	0,47594	0,017144	0,006506	-0,28007	0,410422
tart	0,503411	-0,42771	0,364882	0,284431	0,413183	0,876294	-0,77823	-0,03463	1	0,005314	0,297915	0,124627	-0,34469	0,50577	-0,28134	0,184733	-0,42067
lactic	0,137649	-0,22829	0,281884	0,316156	0,251368	-0,21379	0,177861	0,013902	0,005314	1	-0,49252	0,36133	-0,26212	0,074772	0,208715	0,211808	-0,2313
malic	0,227045	-0,40724	0,353821	0,358508	0,334941	0,654455	-0,40705	0,106449	0,297915	-0,49252	1	0,241601	-0,3836	0,757146	0,375979	0,299114	-0,40868
citric	0,386294	-0,80238	0,831378	0,908058	0,750936	0,132927	-0,35288	-0,55426	0,124627	0,36133	0,241601	1	-0,90471	0,553951	0,554734	0,67559	-0,81306
glyc	-0,59634	0,945742	-0,94465	-0,96466	-0,90524	-0,32745	0,47701	0,47594	-0,34469	-0,26212	-0,3836	-0,90471	1	-0,69696	-0,56528	-0,58961	0,950716
extract	0,565627	-0,77568	0,747124	0,7362	0,732667	0,664446	-0,45428	0,017144	0,50577	0,074772	0,757146	0,553951	-0,69696	1	0,543654	0,556302	-0,77545
NOPA	0,210017	-0,49513	0,521371	0,581047	0,462903	-0,13778	0,250382	0,006506	-0,28134	0,208715	0,375979	0,554734	-0,56528	0,543654	1	0,542706	-0,50223
NH4	0,351867	-0,55362	0,566589	0,621769	0,510025	0,179805	-0,2	-0,28007	0,184733	0,211808	0,299114	0,67559	-0,58961	0,556302	0,542706	1	-0,56001
alc_%	-0,73365	0,999783	-0,98866	-0,96977	-0,97802	-0,39832	0,519319	0,410422	-0,42067	-0,2313	-0,40868	-0,81306	0,950716	-0,77545	-0,50223	-0,56001	1

Figure 14. Correlation plot of showing the correlations of the Fourier Transformed Infrared Spectroscopy (FTIR) data against itself. Colours are used to visually aid the degree of correlation.

Table 7. Alpha diversity statistics on the sterile vineyard 2 fermentations. Includes the Shannon, Ace and Chao1 diversity, Simpson's Evenness, Gini Dominance and Goods Coverage. The order of fermentation stages should be ordered as: Press, Sedimantation (sed), Onset (on), 2.5, 4, 6, 8, 11, rack. The numbers indicate at which alcohol percentage the sample was taken. The names themselves are written as: Timepoint in series - sample series - - sample fermentation stage - - date of measurement, day-month in 2016.

samples	%unclassified	ShannonDiversity	AceDiversity	Chao1Diversity	SimpsonEveness	GiniDominance	GoodsCoverage
10-02a--rack--24-10	26,33215028	2,056215131	4074,169059	4147,5	0,617782595	0,997868637	0,999969431
1-02a--press--07-10	73,82757828	4,076082261	5411,881046	5513,599156	0,803370961	0,989184819	0,999979649
1-02b--sed--08-10	64,36950293	3,479400041	4951,265371	5007,888554	0,749959643	0,992524165	0,999952599
1-02c--on--11-10	33,07321589	2,290555843	4805,860779	4848,254902	0,569550178	0,996748434	0,999994218
2-02b--on--11-10	37,7265419	2,465871616	4675,934943	4806,725076	0,644202069	0,996546162	0,999961819
2-02b--press--07-10	72,7450209	4,309959132	4624,431798	4670,970588	0,808231912	0,988393945	0,999643
2-02c--2-5--13-10	36,23569747	0,890849342	3854,134154	3925,106936	0,168702525	0,998861279	0,999980809
3-02a--sed--08-10	68,18354328	4,308749174	5040,128341	5076,641304	0,857991434	0,989208172	0,999987754
3-02b--2-5--13-10	36,22420102	0,755578533	3908,455386	3915,625571	0,134773655	0,998893508	0,999959489
3-02c--4-0--14-10	35,49156401	0,592489955	3311,674922	3381,07377	0,102748275	0,999230214	0,999959655
4-02a--on--11-10	30,45246242	1,380651019	4403,156406	4510,716463	0,320269333	0,997934179	0,999969867
4-02b--4-0--14-10	34,29809065	0,560061732	2769,376361	2811,744409	0,095704669	0,999254204	0,999944709
4-02c--6-0--15-10	35,121413	0,586647518	2768,032839	2718,043988	0,10134294	0,999243411	0,999878756
5-02a--2-5--13-10	34,73562011	0,763019912	4006,188276	4103,777429	0,142177258	0,998977349	0,999985831
5-02b--6-0--15-10	34,43229812	0,583948707	2841,602056	2840,920548	0,099852371	0,999152338	0,999913662
5-02c--8--17-10	29,02980206	0,493194077	3236,547892	3263,254098	0,082929107	0,999292723	0,999978262
6-02a--4-0--14-10	34,75976652	0,673078536	3627,311609	3655,411192	0,123433189	0,999123135	0,999966156
6-02b--8--17-10	32,28157488	0,535246546	3518,103097	3645,879699	0,089083019	0,999156604	0,99998632
6-02c--11--20-10	30,4261607	0,967084028	4044,588147	4109,909091	0,207958379	0,99880675	0,999985112
7-02a--6-0--15-10	29,61424936	0,60607393	3582,728187	3610,951338	0,109126812	0,999249834	0,99997164
7-02b--11--20-10	28,00830765	0,696090859	4508,115955	4611,53406	0,156277704	0,999252665	0,999992569
7-02c--rack--24-10	20,28115503	1,870080679	3001,272777	3250,035088	0,604074803	0,998885577	0,999975848
8-02a--8--17-10	29,81753427	0,464756977	3701,462327	3766	0,078586477	0,999328897	0,999988869
8-02b--rack--24-10	16,99138595	1,396882112	3650,067094	3774,976096	0,401744853	0,998786342	0,999982862
9-02a--11--20-10	29,55147911	0,806495887	4121,21829	4180,311047	0,150559744	0,998819122	0,999985916

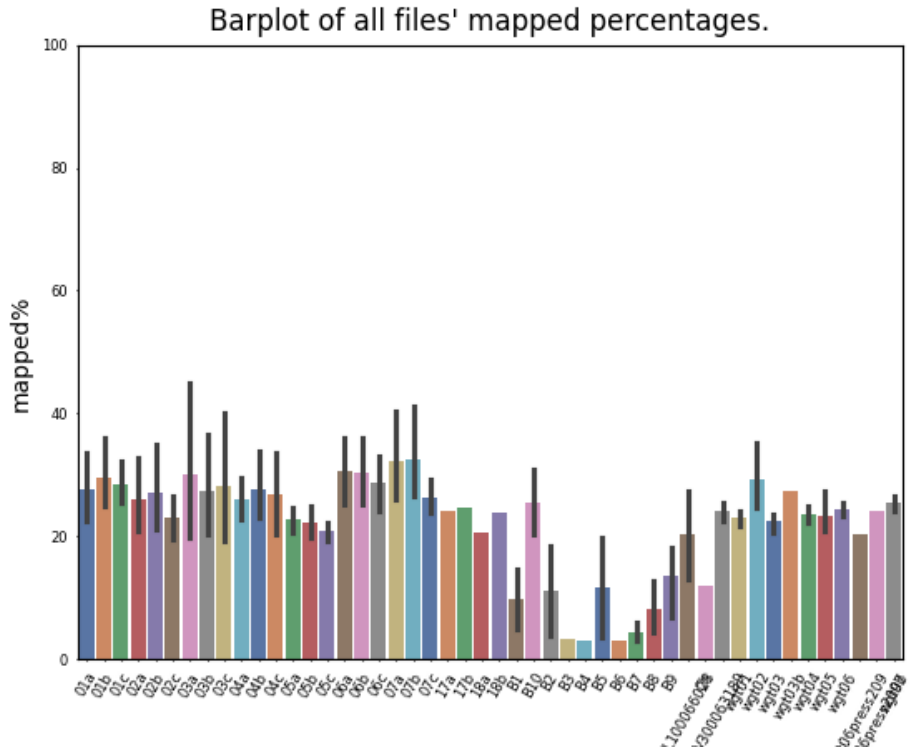


Figure 15. Plot showing per group of samples the percentage of reads mapped to reference and subsequently removed from the metagenomics samples. Includes boxplot of spread excluding outliers for the samples. More specific version of Figure 8 A.

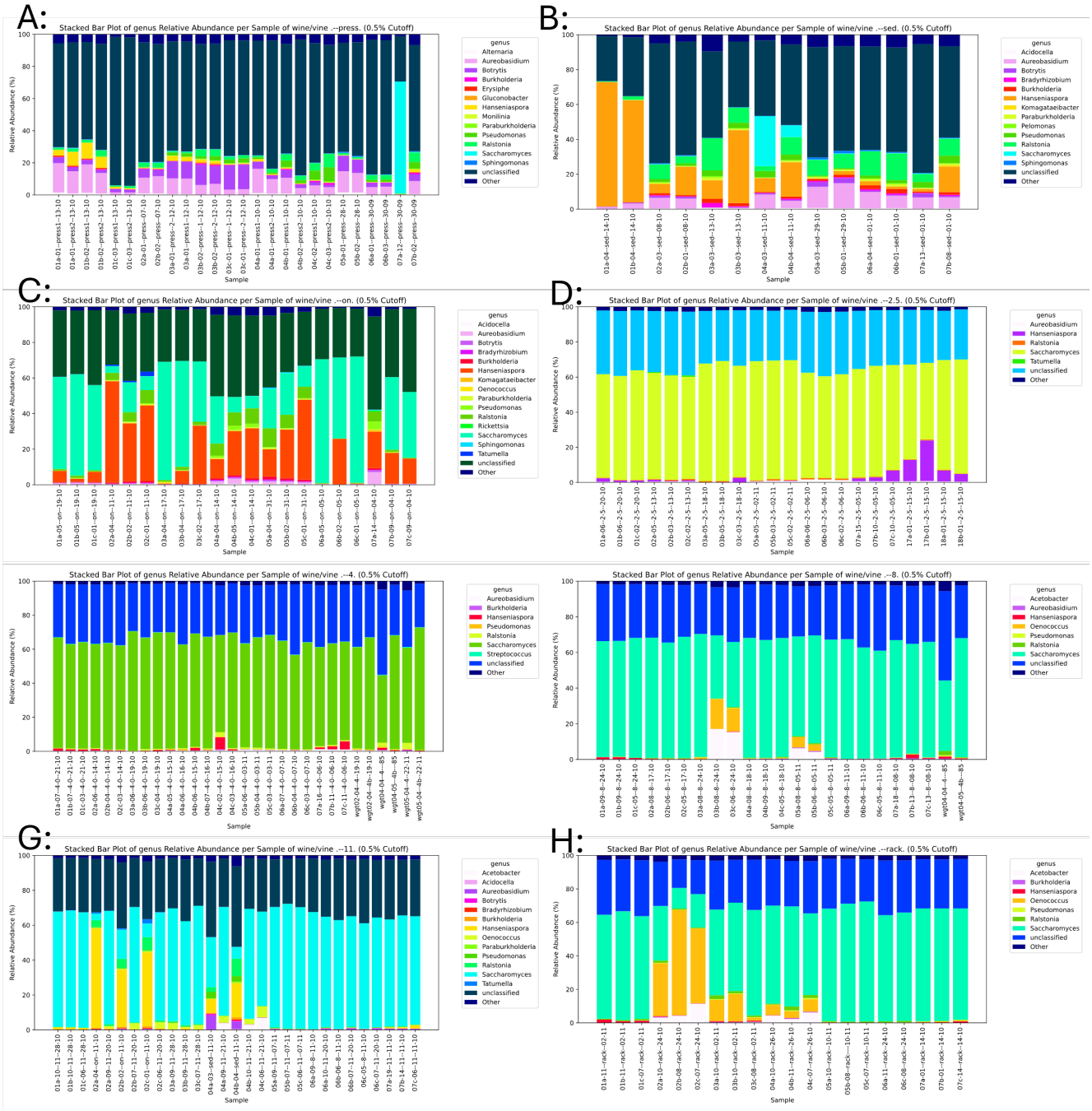
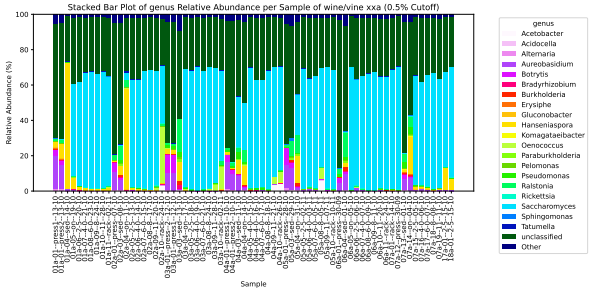
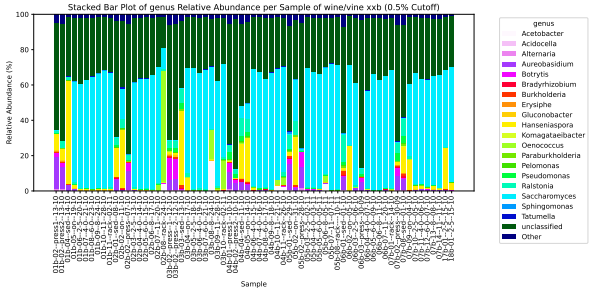


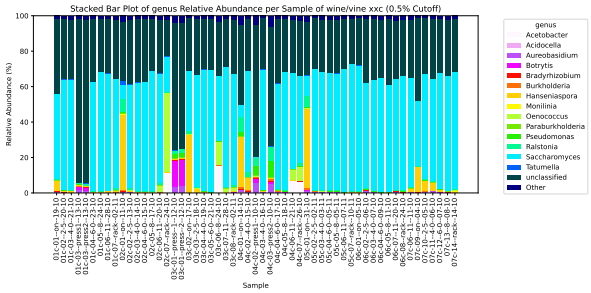
Figure 16. Fermentations shown at different timepoints, only sterile as they have the measurement times noted. (**A**) Press. (**B**) Sedimentation. (**C**) Fermentation onset. (**D**) 2.5% alcohol. (**E**) 4% alcohol. (**F**) 8% alcohol. (**G**) 11% alcohol. (**H**) Rack phase.



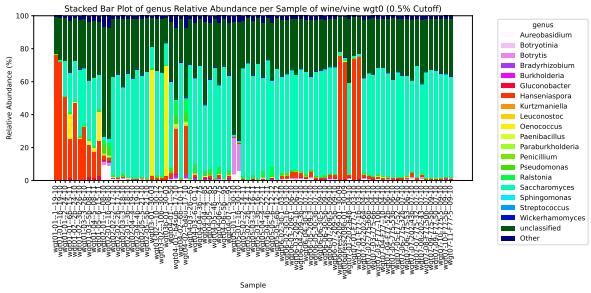
(a) ***All sterile bioreplicate A fermentations visualized.***



(b) ***All sterile bioreplicate B fermentations visualized.***



(c) ***All sterile bioreplicate C fermentations visualized.***



(d) ***All winery fermentations visualized.***

Figure 17. Visualizations of all sterile bioreplicate fermentations and winery fermentations. **Understanding sample names:** The fermentations are not always perfectly sorted due to file naming issues. The order of fermentation stages is: Press, Sedimentation (sed), Onset (on), 2.5, 4, 6, 8, 11, rack. The numbers indicate the alcohol percentage at which the sample was taken. If there is less than 0.5 percent abundance, the genus is placed under 'other'.

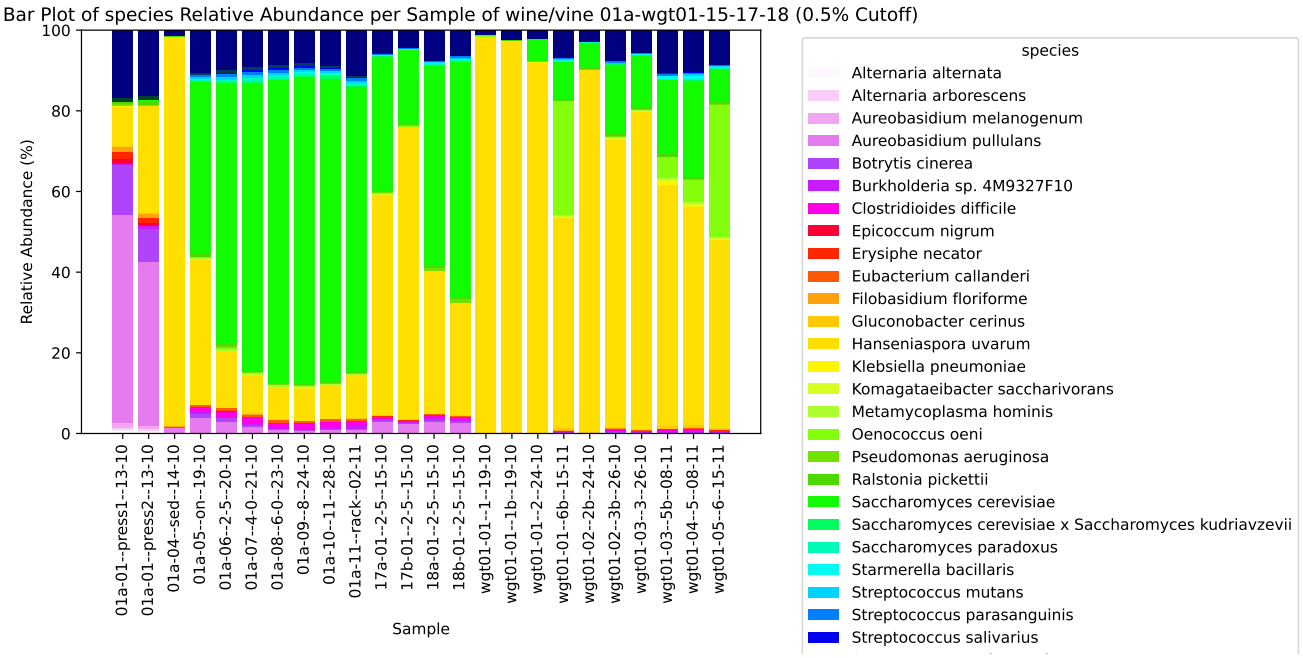


Figure 18. Vineyard and winery 1 data shown on species level. Shows same samples as Figure 9 B.

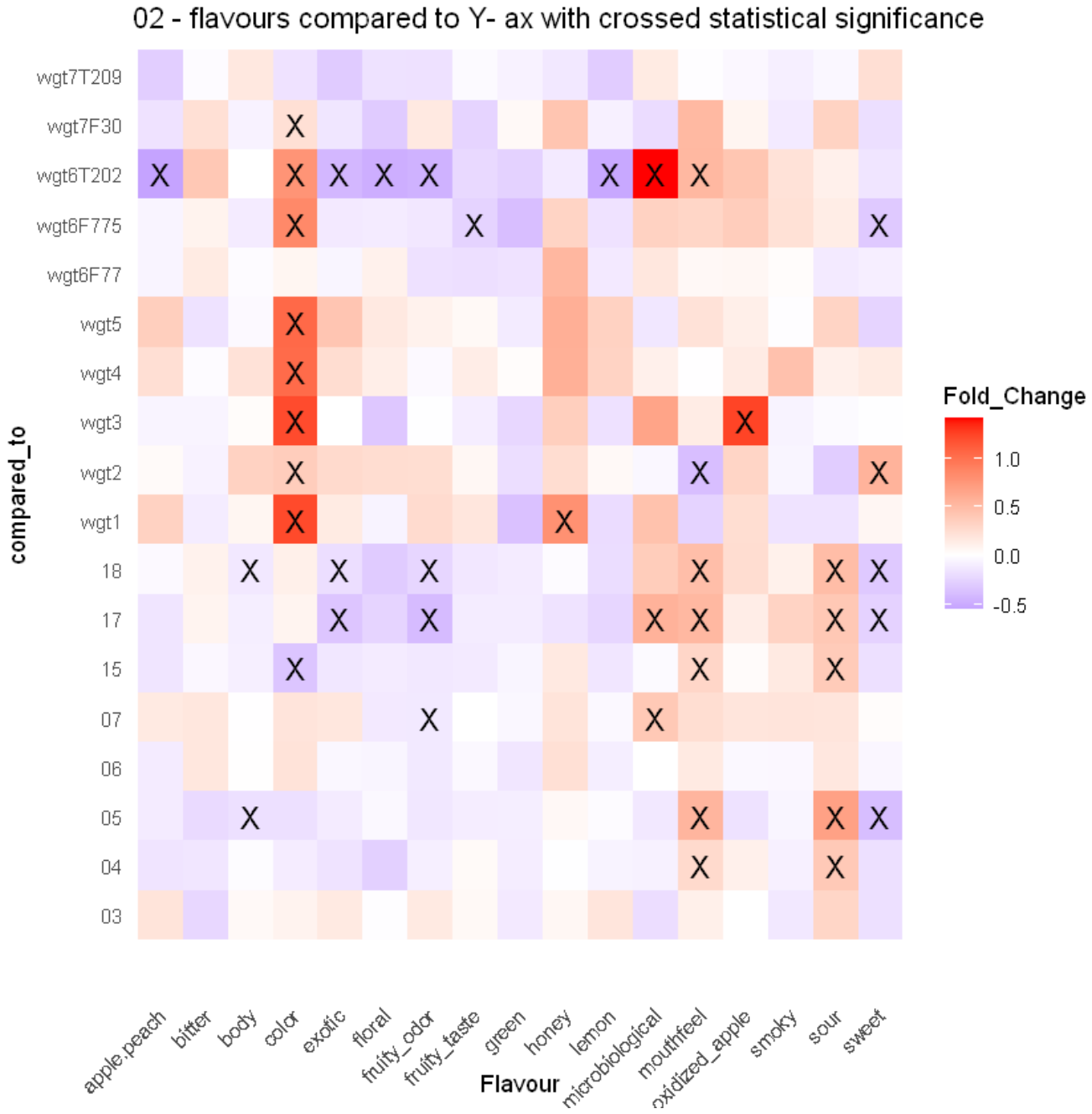


Figure 19. ***Relative change of sensory values with significance marked with an X over all sensory values. Uses merged biological replicates of vineyard 2 as base for comparison.*** The figure contains uses biological replicates from sterile fermentation vineyard 1 as base, having calculated the relative change as compared to base. ANOVA with Tukey multiple testing correction was utilized to find significant differences, marked with an 'X' in the figure.

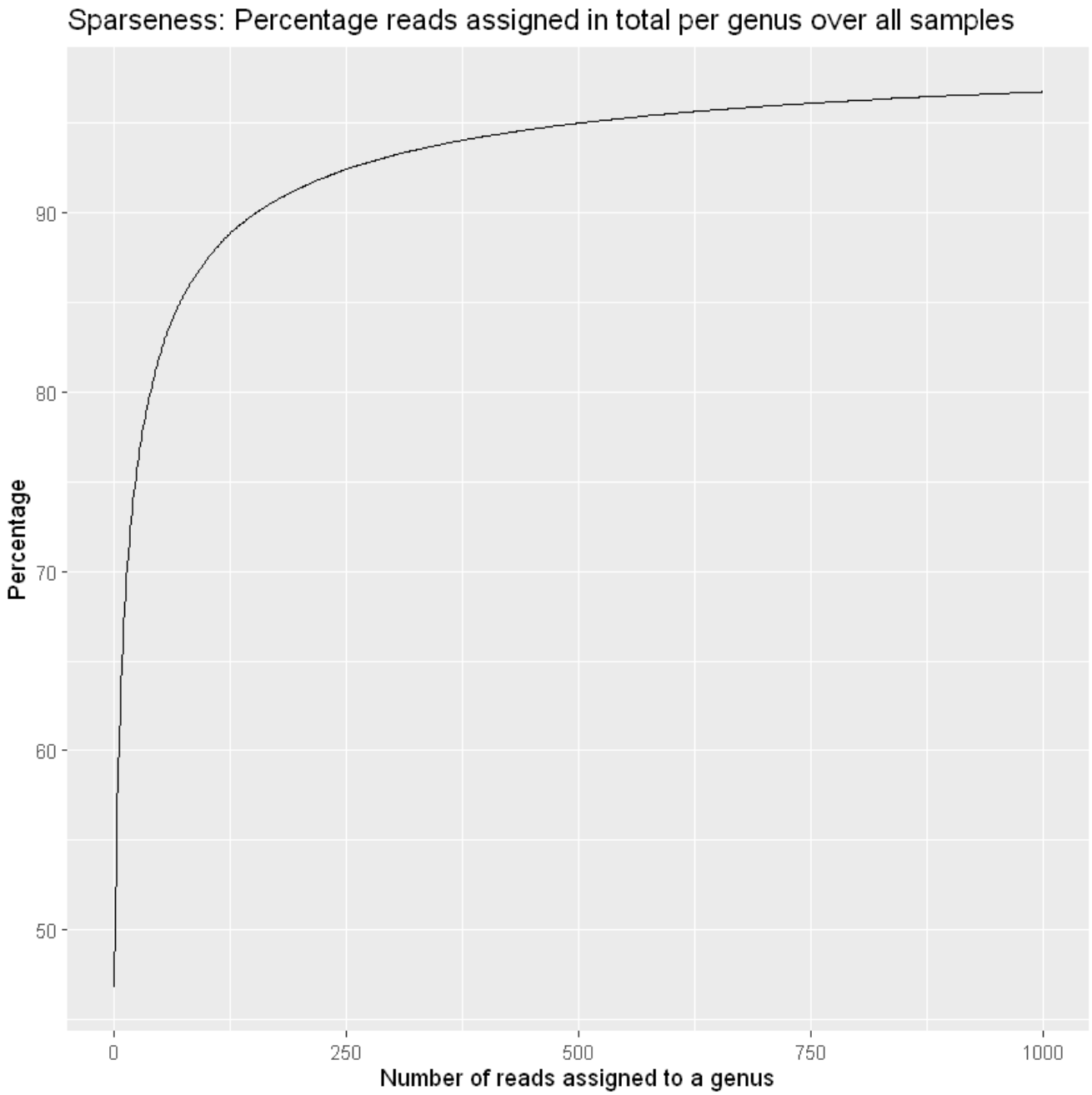


Figure 20. ***Fermentation onset data is very sparse.*** **(a) Fermentation onset data is sparse, with 43 percent of all cells containing a value of 0. There is 62 percent of reads with the value of 5 or lower.**

Table 8. The top 25 Spearman correlating factors to sensory properties, sorted by P- value.

Data_Origin	Factor	flavour	correlation	sum_factor	sum_reads	p_value	adjusted_p_value
micobes_start	Monilinia_38447	mouthfeel	0,863157895	-0,073488936	26639	0	0
micobes_start	Sclerotinia_5179	mouthfeel	0,867669173	-0,118959706	22116	0	0
ferm_start	total_acid	sour	0,855639098	189,33	189,33	0	0
majors	maj_Isoamyl_alcohol	bitter	0,882706767	2425,803933	2425,803933	0	0
ferm_fin	fin_tart	sour	0,886047449	70,06	70,06	2,02E-07	0,010199181
micobes_start	Botryotinia_40558	mouthfeel	0,893233083	-0,026849891	32936	3,12E-07	0,013113357
ferm_fin	fin_pH	sour	-0,875569595	57,5	57,5	4,29E-07	0,015453413
ferm_fin	fin_total_acid	sour	0,871004198	149,54	149,54	5,83E-07	0,018379759
micobes_start	Aureobasidium_5579	mouthfeel	0,82406015	58,58131383	5136237	3,84E-06	0,096473784
micobes_start	Epicoccum_104397	mouthfeel	0,82406015	-0,207748994	14588	3,84E-06	0,096473784
microbes_finish	Pedobacter_84567	sour	0,923308271	-0,231199131	15510	4,21E-06	0,096473784
ferm_fin	fin_total_acid	sweet	-0,83145222	149,54	149,54	5,56E-06	0,116861455
ferm_start	extract	sour	0,818045113	456,66	456,66	7,84E-06	0,118177393
minors	min_Geraniol	sour	-0,812030075	151,3124131	151,3124131	1,31E-05	0,118177393
micobes_start	Botrytis_33196	mouthfeel	0,810526316	11,73295546	1196051	1,46E-05	0,118177393
ferm_fin	fin_malic	sour	0,80812664	51,47	51,47	1,63E-05	0,118177393
ferm_fin	fin_tart	sweet	-0,804364184	70,06	70,06	1,91E-05	0,118177393
ferm_start	pH	body	0,800303477	52,13	52,13	2,26E-05	0,118177393
ferm_fin	fin_malic	bitter	-0,793077726	51,47	51,47	3,03E-05	0,118177393
micobes_start	Alternaria_5598	color	-0,798496241	1,359783819	159025	3,08E-05	0,118177393
ferm_start	pH	sour	-0,790506918	52,13	52,13	3,35E-05	0,118177393
majors	maj_Diethyl_succinate	bitter	0,790977444	8,536687988	8,536687988	4,49E-05	0,118177393
microbes_finish	Buchnera_32199	fruity_odor	-0,787969925	-0,270954874	1364	5,17E-05	0,118177393
micobes_start	Alternaria_5598	mouthfeel	0,786466165	1,359783819	159025	5,53E-05	0,118177393

Table 9. Correlations to Tenebriocola, a beetle. False positives.

Data_Origin	Factor	flavour	correlation	sum_factor	sum_reads	p_value	adjusted_p_value
microbes_finish	Tenebrionicola_2943312	mouthfeel	0,640601504	-0,277473303	10	0,002942956	0,118177393
micobes_start	Tenebrionicola_2943312	exotic	0,590977444	-0,343687284	151	0,007060087	0,130205275
microbes_finish	Tenebrionicola_2943312	fruity_odor	-0,563909774	-0,277473303	10	0,01080682	0,136497296
microbes_finish	Tenebrionicola_2943312	body	-0,493233083	-0,277473303	10	0,028661115	0,220904121
microbes_finish	Tenebrionicola_2943312	sour	0,484210526	-0,277473303	10	0,03206396	0,220904121
microbes_finish	Tenebrionicola_2943312	fruity_taste	-0,452631579	-0,277473303	10	0,046575392	0,249536651

Table 10. All Spearman correlations related to Botrytis with a $P < 0.05$.

Data_Origin	Factor	flavour	correlation	sum_factor	sum_reads	p_value	adjusted_p_value
microbes_finish	Botrytis_33196	apple,peach	-0,457142857	0,301573172	152521	0,044234785	0,249536651
micobes_start	Botrytis_33196	apple,peach	-0,553383459	11,73295546	1196051	0,0126451	0,149173284
micobes_start	Botrytis_33196	body	-0,520300752	11,73295546	1196051	0,020148181	0,219806887
micobes_start	Botrytis_33196	color	-0,711278195	11,73295546	1196051	0,00063059	0,118177393
micobes_start	Botrytis_33196	exotic	-0,527819549	11,73295546	1196051	0,018188281	0,202889946
microbes_finish	Botrytis_33196	floral	-0,469172932	0,301573172	152521	0,038442746	0,249536651
micobes_start	Botrytis_33196	fruity_odor	-0,539849624	11,73295546	1196051	0,015375697	0,175893127
microbes_finish	Botrytis_33196	fruity_odor	-0,616541353	0,301573172	152521	0,004574385	0,130205275
microbes_finish	Botrytis_33196	green	0,45112782	0,301573172	152521	0,04737678	0,249536651
microbes_finish	Botrytis_33196	microbiological	0,460150376	0,301573172	152521	0,042726403	0,249536651
micobes_start	Botrytis_33196	mouthfeel	0,810526316	11,73295546	1196051	1,46E-05	0,118177393
microbes_finish	Botrytis_33196	smoky	0,492666451	0,301573172	152521	0,027318199	0,220904121
micobes_start	Botrytis_33196	sour	0,612030075	11,73295546	1196051	0,004950589	0,130205275

Table 11. All Spearman correlations related to several organisms or groups with more or less interest with a P < 0.05.

Data_Origin	Factor	flavour	correlation	sum_factor	sum_reads	p_value	adjusted_p_value
micobes_start	Tatumella_82986	sweet	0,604738666	8,019809501	2109345	0,004733177	0,130205275
micobes_start	Tatumella_82986	sour	-0,768421053	8,019809501	2109345	0,00011412	0,118177393
micobes_start	Tatumella_82986	bitter	0,673684211	8,019809501	2109345	0,001508472	0,118177393
micobes_start	Tatumella_82986	exotic	0,601503759	8,019809501	2109345	0,005928434	0,130205275
microbes_finish	Tatumella_82986	honey	-0,476298103	2,059345534	510416	0,033751518	0,229463383
microbes_finish	_10239	sour	-0,557894737	0,087381837	89985	0,011828318	0,141061072
microbes_finish	_10239	honey	-0,504891038	0,087381837	89985	0,02317699	0,220904121
microbes_finish	_10239	smoky	0,493418613	0,087381837	89985	0,027047673	0,220904121
microbes_finish	_10239	sweet	0,667168155	0,087381837	89985	0,001311753	0,118177393
micobes_start	Hanseniaspora_29832	green	0,536842105	860,4729929	142333667	0,016043222	0,182635226
microbes_finish	Hanseniaspora_29832	exotic	0,526315789	11,99065955	4397086	0,018567365	0,206434221
microbes_finish	Saccharomyces_4930	mouthfeel	0,669172932	1561,906702	570844489	0,001660346	0,118177393
microbes_finish	Saccharomyces_4930	sour	0,535338346	1561,906702	570844489	0,016385702	0,186089075
microbes_finish	Saccharomyces_4930	fruity_taste	-0,457142857	1561,906702	570844489	0,044234785	0,249536651
microbes_finish	Saccharomyces_4930	body	-0,533834586	1561,906702	570844489	0,016734094	0,189270543
micobes_start	Saccharomyces_4930	green	-0,541353383	1142,956113	236200013	0,015050505	0,172626918
microbes_finish	Saccharomyces_4930	fruity_odor	-0,637593985	1561,906702	570844489	0,003115674	0,118177393
microbes_finish	Burkholderia_32008	fruity_odor	0,595488722	6,105356725	2337523	0,006555127	0,130205275
microbes_finish	Paraburkholderia_1822464	fruity_odor	0,565413534	2,512621499	1027597	0,010563067	0,136497296
micobes_start	Burkholderia_32008	green	0,518796992	26,06384712	3207250	0,020559898	0,220904121
microbes_finish	Burkholderia_32008	lemon	0,503759398	6,105356725	2337523	0,02506256	0,220904121
micobes_start	Paraburkholderia_1822464	green	0,488721805	15,72194443	1953613	0,030324491	0,220904121
microbes_finish	Paraburkholderia_1822464	lemon	0,470676692	2,512621499	1027597	0,037763045	0,249536651
micobes_start	Burkholderia_32008	sour	0,458646617	26,06384712	3207250	0,043475454	0,249536651
micobes_start	Burkholderia_32008	color	-0,460150376	26,06384712	3207250	0,042726403	0,249536651
microbes_finish	Paraburkholderia_1822464	mouthfeel	-0,460150376	2,512621499	1027597	0,042726403	0,249536651
micobes_start	Paraburkholderia_1822464	color	-0,494736842	15,72194443	1953613	0,028123156	0,220904121
microbes_finish	Paraburkholderia_1822464	microbiological	-0,536842105	2,512621499	1027597	0,016043222	0,182635226
microbes_finish	Burkholderia_32008	microbiological	-0,557894737	6,105356725	2337523	0,011828318	0,141061072
micobes_start	Paraburkholderia_1822464	exotic	-0,606015038	15,72194443	1953613	0,005491534	0,130205275
micobes_start	Burkholderia_32008	exotic	-0,633082707	26,06384712	3207250	0,003390368	0,124011085
micobes_start	Ralstonia_48736	color	-0,506766917	196,6541215	23882879	0,024103986	0,220904121
micobes_start	Ralstonia_48736	exotic	-0,606015038	196,6541215	23882879	0,005491534	0,130205275
microbes_finish	Ralstonia_48736	fruity_odor	0,532330827	26,30388675	10246996	0,017088471	0,192691433
micobes_start	Ralstonia_48736	green	0,485714286	196,6541215	23882879	0,031475535	0,220904121
microbes_finish	Ralstonia_48736	microbiological	-0,551879699	26,30388675	10246996	0,01292737	0,151111538
microbes_finish	Ralstonia_48736	mouthfeel	-0,47518797	26,30388675	10246996	0,0357808	0,239668019
microbes_finish	Ralstonia_48736	smoky	-0,478375363	26,30388675	10246996	0,032875213	0,224598484
micobes_start	Ralstonia_48736	sour	0,457142857	196,6541215	23882879	0,044234785	0,249536651

Table 12. All Spearman correlations related to Aureobasidium with a P < 0.05.

Data_Origin	Factor	flavour	correlation	sum_factor	sum_reads	p_value	adjusted_p_value
micobes_start	Aureobasidium_5579	apple,peach	-0,62406015	58,58131383	5136237	0,003999881	0,130205275
microbes_finish	Aureobasidium_5579	apple,peach	-0,512781955	6,021341316	1709554	0,022275223	0,220904121
microbes_finish	Aureobasidium_5579	bitter	0,44962406	6,021341316	1709554	0,048188917	0,249536651
micobes_start	Aureobasidium_5579	body	-0,568421053	58,58131383	5136237	0,010089051	0,136497296
micobes_start	Aureobasidium_5579	color	-0,669172932	58,58131383	5136237	0,001660346	0,118177393
micobes_start	Aureobasidium_5579	exotic	-0,57443609	58,58131383	5136237	0,009193214	0,136497296
microbes_finish	Aureobasidium_5579	fruity_odor	-0,603007519	6,021341316	1709554	0,005779732	0,130205275
micobes_start	Aureobasidium_5579	fruity_odor	-0,581954887	58,58131383	5136237	0,008166199	0,136497296
micobes_start	Aureobasidium_5579	fruity_taste	-0,491729323	58,58131383	5136237	0,029207266	0,220904121
microbes_finish	Aureobasidium_5579	lemon	-0,488721805	6,021341316	1709554	0,030324491	0,220904121
micobes_start	Aureobasidium_5579	mouthfeel	0,82406015	58,58131383	5136237	3,84E-06	0,096473784
microbes_finish	Aureobasidium_5579	mouthfeel	0,452631579	6,021341316	1709554	0,046575392	0,249536651
micobes_start	Aureobasidium_5579	sour	0,616541353	58,58131383	5136237	0,004574385	0,130205275